

Degree Project in Technology, Work and Health
Second cycle, 30 credits

Evaluating financial effects of the work environment

Recent research and current practices in a selection of Swedish organisations

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TRITA – CBH-GRU-2026:002
Stockholm, Sweden 2026

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26/01/09

Master's Thesis

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Abstract

Work-related ill-health poses a significant financial burden on individuals, organisations, and society. This thesis aims to enhance understanding of work environment economics by reviewing recent research and examining how selected Swedish organisations in manufacturing, logistics, healthcare and higher education address the financial effects of the work environment.

The literature review, covering studies from 2013 to 2024, showed that general business and health economic methods have taken precedence over targeted work environment economics approaches in recent scientific literature. An interview study with five organisations revealed high awareness but limited use of work environment economics tools. Participants expressed a need for simpler yet credible methods that can be adapted to local business contexts. Alternative approaches emphasised proactive strategies and the integration of systematic work environment management into daily operations.

Motives for taking work environment effects into account included occupational safety and health professionals raising awareness among management about the benefits of systematic work environment management and occupational health interventions. The findings suggest that work environment economics calculations may be especially useful in organisations with a lower maturity in systematic work environment management, where financial arguments are more critical for justifying interventions. In contrast, organisations with a strong integration of systematic work environment economics into core operations rely less on formal economic arguments, as the value of a healthy work environment is already embedded in their culture and organisational goals.

Based on the results of the interview study, it is concluded that future developers of work environment economics tools should prioritise participatory design and practical usability to ensure relevance and effectiveness in real-world settings. In combination with proactive approaches and the integration of systematic work environment management into business operations, work environment economic tools can support strategic decision-making and ethical, sustainable investments in employee health and well-being.

Keywords

work environment, work environment economics, health economics, ergonomics, human factors, systematic work environment management, organisational learning

Sammanfattning

Arbetsrelaterad ohälsa innebär betydande ekonomiska konsekvenser för individer, organisationer och samhället. Syftet med detta examensarbete är att öka förståelsen för arbetsmiljöekonomi genom att granska aktuell forskning och undersöka hur ett urval av svenska organisationer inom tillverkning, logistik, vård och högre utbildning tar hänsyn till de ekonomiska effekterna av arbetsmiljön.

Litteraturöversikten, som omfattar studier från 2013 till 2024, visar att generella företagsekonomiska och hälsoekonomiska metoder har fått företräde framför mer specifika arbetsmiljöekonomiska verktyg i den senaste vetenskapliga litteraturen. En intervjustudie med fem organisationer visade att medvetenheten om arbetsmiljöekonomiska verktyg är hög, men att användningen är begränsad. Deltagarna uttryckte ett behov av enklare men samtidigt trovärdiga metoder som kan anpassas till lokala verksamhetsförhållanden. Framstående alternativa angreppssätt till användning av arbetsmiljöekonomiska metoder som identifierades var proaktivitet och integration av systematiskt arbetsmiljöarbete i den dagliga verksamheten.

Motiv till att beakta arbetsmiljöns ekonomiska effekter inkluderade arbetsmiljöspecialisters arbete med att öka ledningens medvetenhet om nyttan av systematiskt arbetsmiljöarbete arbetsmiljöförbättrande insatser. Resultaten tyder på att ekonomiska kalkyler kan vara särskilt värdefulla i organisationer med låg mognadsgrad i det systematiska arbetsmiljöarbetet, där finansiella argument ofta krävs för att motivera åtgärder. I kontrast förlitar sig organisationer med hög grad av integrering av arbetsmiljö i kärnverksamheten mindre på formella ekonomiska argument, eftersom värdet av en god arbetsmiljö redan är förankrat i deras kultur och mål.

Slutsatsen som kan dras utifrån resultatet av intervjustudien är att framtida utvecklare av arbetsmiljöekonomiska verktyg bör prioritera deltagardriven design och praktisk användbarhet för att säkerställa relevans och effektivitet i den löpande verksamheten. I kombination med proaktiva tillvägagångssätt och integration av det systematiska arbetsmiljöarbetet i verksamhetsstyrningen, kan arbetsmiljöekonomiska metoder stödja strategiskt beslutsfattande och främja etiska och hållbara investeringar i arbetsmiljön.

Nyckelord

arbetsmiljö, arbetsmiljöekonomi, hälsoekonomi, ergonomi, systematiskt arbetsmiljöarbete, organisatoriskt lärande

Acknowledgments

I would like to thank the following people for having supported this work:

Linda Rose, my supervisor. Thanks for your invaluable guidance and encouragement!

Andrea Eriksson. Your supervision on the qualitative methods was very helpful!

Malin Lohela Karlsson, for generously sharing your time and expertise within the field.

Emmanuel Aboagye, for helpful guidance in what work environment economics is truly about.

All interviewees - without your time and engagement this work had been impossible.

Pontus Nordgren for volunteering in the pilot interview, as well as keeping up a helpful study spirit together with Christina Papadimou in the Pomodoro Sessions.

Henrik Bäcklin & Sofie Wettergård for helping with recruitment of interviewees.

Emilia Asplund & Julia McTaggart for your highly appreciated companionship during the writing process.

Erik, for your unwavering emotional and academic support.

Miranda, Elvira & Ivar: Love you to the moon and back.

Stockholm, January 2026
Mia Bäcklin

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List of key terms, acronyms and abbreviations

Term	Explanation	More information
Direct costs	Costs that can be clearly and directly attributed to a specific product, service, project or department, e.g. raw materials in manufacturing, salaries.	2.5.2; Johansson & Johrén (2017)
Economic evaluation	A systematic method for comparing the costs and consequences (outcomes) of two or more alternatives, to support evidence-based decision-making.	Drummond (2015)
Economic methods	Tools, techniques, or approaches used to collect, analyse, and interpret economic data.	Drummond (2015)
Economic models	Structured representations (mathematical, statistical, or conceptual) of reality, used to simulate, compare, or predict economic outcomes over time.	Drummond (2015)
Friction costs	Costs that arise from employee abseentism or turnover	2.5.6; Koopmanschap et al. (1995); Johansson & Johrén (2017)
HE	Health Economics	2.4
HEE	Health Economic Evaluation	2.4
Hidden costs	Harder to quantify-costs that arise as a consequence of poor work environments. Do not always appear in formal accounting but can have a significant financial impact. E.g. reduced productivity due to fatigue, stress, or poor ergonomics; Decreased work quality or increased error rates, friction costs.	2.5.1; Johansson & Johrén (2017)
In-direct costs	Costs that cannot be directly traced to one specific product or person, e.g. Overhead costs, IT infrastructure.	2.5.2; Johansson & Johrén (2017)

KPI	Key Performance Indicator	2.6
OHS	Occupational Health Services	
OSH	Occupational Safety and Health	
RCT	Randomized Controlled Trial: A type of scientific study used to test the effectiveness of a treatment, intervention, or program. Participants are randomly assigned to different groups, typically one treatment group and one control group.	(Matthews, 2006)
SWEA	Swedish Work Environment Authority	Arbetsmiljöverket (n.d.)
SWEM	Systematic Work Environment Management	2.1
Visible costs	Easily measurable costs that appear clearly in financial records. They are typically acknowledged by organisations and are often included in cost analyses. E.g. salary, sick leave compensation, legal costs from workplace incidents	2.5.1; Johansson & Johrén (2017)
WE	Work Environment	2.1
WEE	Work environment economics	2.3

List of economic methods

Method	Description	Formula	More information
Annuity method	A method used to calculate the present value of a stream of equal payments made at regular intervals.	PV = PMT × [1 - (1 + r)^-n] / r Where: PV = Present Value PMT = Payment per period r = Interest rate per period n = Number of periods	Drummond (2015)
Cost-benefit analysis (CBA)	A method where the benefits and costs of a project or decision are expressed in monetary terms.	Cost-benefit analysis ratio = Present value of benefit/Present value of cost	Drummond (2015)
Cost-effectiveness analysis (CEA)	Compares the relative costs and outcomes (effects) of two or more interventions.	ICER = (C1 - C0) / (E1 - E0) Where: C1 = Cost of intervention 1 C0 = Cost of intervention 0 (comparator) E1 = Effect of intervention 1 E0 = Effect of intervention 0	Drummond (2015)
Cost-utility analysis (CUA)	Similar to CEA but uses utility-based measures (e.g., QALYs) for outcomes.	ICUR = (C1 - C0) / (U1 - U0) Where: C1 = Cost of intervention 1 C0 = Cost of intervention 0 (comparator) U1 = Utility (e.g., QALYs) of intervention 1 U0 = Utility of intervention 0	Drummond (2015)
Final value method	Calculates the future value of an investment after a certain period.	FV = PV × (1 + r)^n Where: FV = Future Value PV = Present Value r = Interest rate per period n = Number of periods	Drummond (2015)

The friction cost method (FCM)	Estimates productivity losses during the *friction period,* due to illness, disability, or premature death.	$Productivity\ Loss =$ $(W * D) * 1/((1 + r)\ 2\text{^}\ )$ Where: W = Wage rate or value of output per day (or hour) D = Duration of the friction period (in days or hours) r = Discount rate (for adjusting to present value) t = Time (usually 1 if all loss occurs within the first year)	Koopmansch ap et al., (1995)
The human capital method (HCM)	Estimates the economic impact of disease or injury in terms of lost productivity. The value of lost productivity due to sick leave, disability or premature death can be calculated.	$Productivity\ Loss = \sum_{t=1}^{T} (W_t * L_t) * \frac{1}{(1+r)^t}$ Where: W_t = Wage or income at time *t* L_t = Proportion of lost productivity (e.g. 1 for full time, 0.5 for part time) *T*= Total number of periods (e.g. months or years of absence until retirement or death) *r* = Discount rate $\frac{1}{(1+r)^t}$ = Discount factor for year *t*	Drummond (2015); Pike & Grosse (2018)
Pay back method	Determines the length of time required to recover the cost of an investment.	Payback Period = Initial Investment / Average Annual Cash Inflow	Drummond (2015)
Present value method	Calculates the present value of future cash flows.	PV = Σ [CFt / (1 + r)^t] Where: CFt = Cash flow at time t	Drummond (2015)

		r = Discount rate t = Time period	
Return on investment (ROI)	A measure of the profitability of an investment.	ROI = (Current value of investment – Cost of investment) /Cost of Investment	Drummond (2015)
Systematic accident cost analysis method (SACA)	A guide in identifying the activities generated by a work injury and calculating their costs for the company.	N/A	Popescu-Stelea et al., (2019)

1. Introduction

This chapter introduces the specific problem addressed in this thesis, along with its context and the overall aim of the project. It also outlines the structure of the thesis.

1.1 Work-related ill-health and its economic effects

Work-related ill-health is a significant financial issue for the individual, the organisation and the society. According to The Swedish Work Environment Authority, SWEA (Arbetsmiljöverket, 2023), one in four Swedish employees report health problems due to their job, such as fatigue, body aches or anxiety. Fatigue, physical pain or ache, and musculoskeletal disorders (MSDs) such as neck, shoulder, and arm problems are the most common types of complaints among employees who have problems because of work, and account together for 80 % of reported *health problems caused by work* (Arbetsmiljöverket, 2023), whereas *accidents at work* account for 20 % of the reported health problem cases.

Individual costs of work-related ill-health include income loss due to sick leave, as well as costs for healthcare and medication. Sick leave can also impact career progression and social costs such as lowered quality of life and strain on family relationships as well as reduced participation in social activities. In turn, societal costs due to work-related ill-health include healthcare burden, economic productivity loss and social insurance and welfare costs. On the organisational level, work-related ill-health poses economic burden due to costs for sick leave, operational disruptions, productivity losses due to both presenteeism and absenteeism, and can also cause harm to employer brand, with difficulties attracting and retaining talent as a consequence. (European Agency for Safety and Health at Work, 2013).

Costs due to a poor work environment can be reduced by investments in occupational safety and health (OSH), but resources allocated for OSH interventions are limited within organisations, and the economic value of such interventions often need to be proven to attract employers (Grimani et al., 2018). Investing in employee well-being is often seen as a cost rather than an investment, and as a trade-off with other organisational goals (De Neve, 2023). However, De Neve et al. (2023) reported a strong positive relationship between employees self-rated workplace well-being and firm performance for over 1,600 listed companies in the United States. The firm-level evidence presented in their work suggests that there may be strong business-related reasons to invest in employee well-being (De Neve, 2023).

In Sweden, mental disorders became a more common reason for long-term sick leave between 2010 and 2016 (Skandia, 2024). In 2010, 30 % of sick leave was due to mental illness, while since 2016 has remained stable at around 46 % (Skandia, 2024). Strong evidence of positive economic results for Return-to-Work interventions after an absence due to a common mental disorder, both from the employer's and the societal perspectives, has been found in a systematic review including 11 economic evaluation studies (Gaillard, 2020).

Rose et al. (2007 & 2013) provided an overview as well as an evaluation of models used to link Work Environment (WE) to business results, together with a report of which tools were used in a selection of Swedish organisations at that time (Rose et al., 2007). As time has passed, it is time to follow up this work with an up-to-date report of what models have been reported in research after 2013, and a status update on how a selection of Swedish organisations today link their work environment to their business results.

1.2 Relevance of the thesis project

This thesis project is relevant to the master's degree in Technology, Work and Health, as evidence-based knowledge in the field of work environment economics will provide work environment professionals with a stronger foundation for arguing in favour of occupational safety and health investments. The choice of focus on the organisational perspective, and not the societal or individual economic effects of work environment, is based on the assumption that organisations themselves have the strongest possibilities to influence the work environment of their employees. Demonstrating the financial benefits of a healthy work environment therefore has the potential to enable more investments in workplace conditions, which in turn could make a significant difference to employee well-being. Promoting the well-being of employees, as well as organisational productivity and overall system performance, are objectives of ergonomics according to the International Ergonomics Association (IEA, n.d.)

1.3 Problem

Traditionally, most intervention studies on OSH interventions have focused on interventions' effectiveness rather than cost-effectiveness (Tompa et al., 2009). Although the typical payback time for ergonomic interventions has been found to be less than one year (Goggins, 2008), stating only that an intervention is effective will not be enough, since *all* interventions may not always give a financial return to the organisation (Grimani et al., 2018; Tompa et al., 2009). Knowledge and understanding of work environment economics (WEE) form an important foundation for organisational decision-making, particularly when assessing the costs and benefits of a planned investment for reducing work environment risks, as well as when comparing different investment alternatives for improving the work environment (Grimani et al., 2018). However, Rose & Orrenius (2007) found that WEE methods were only used in a few of 17 Swedish organisations included in their study, and none of the representatives contacted believed that the methods they had access to were satisfactory (Rose & Orrenius, 2007).

1.4 Aim

The aim of this thesis was to contribute to the understanding of WEE by presenting a review of recent research in the field and describing how a selection of Swedish organisations in the manufacturing, logistics, healthcare and higher education sectors take into account and address the financial effects of their work environment.

1.5 Research questions

The research questions of this thesis project were:

1. What economic methods have been described in research from 2012-2024 related to organisational financial impact of work environment risks, problems and interventions?

2. How does a selection of Swedish organisations take into account and approach economic effects of their work environment?

Sub questions:

2a) What motives are there in organisations, for taking WEE effects into account?

b) What WEE methods do representatives from the organisations know about?

c) What WEE methods do the organisations use, if any, or what other approaches to economic effects of the work environment do they have?

d) In what way has the use of WEE methods facilitated the organisations work? What is difficult? What is not?

e) What is on the selected organisations "wish list" in the field of WEE? Can WEE methods be better fitted to organisations needs? What would be needed, for them to use WEE methods?

1.6 Sustainable development

All master theses from KTH Royal Institute of Technology should reflect on the Sustainable Development Goals from the Agenda 2030 (United Nations, 2015). The topic of this thesis reflects on Goal 3: Good Health and Well-being, and contributes especially to target 3.d: *Strengthen the capacity of all countries, in particular developing countries, for early warning, risk reduction and management of national and global health risks*. This thesis also contributes to Goal 8: Decent Work and Economic Growth, especially 8.3 *Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services;* and 8.5: *By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.* Target 8.5 is relevant for this thesis since work-related ill-health is not equally distributed between the sexes. For example, in Sweden 2022, work-related disorders (including both problems caused by accidents and problems caused by other circumstances than accidents) were more common among women (30 %) than among men (21 %). Occupations in healthcare and education, which are dominated by women, had a relatively high proportion of employees with work-related disorders (Arbetsmiljöverket, 2023). Men, on the other hand, suffer more often from fatal accidents at work (Statista, 2025).

1.7 Research methodology

This project was carried out as a scoping literature review and an interview study where representatives from five Swedish organisations were interviewed on their experiences in the field of WEE. Scoping reviews are typically used to identify the types of evidence in a given field, or to identify and analyse knowledge gaps, while systematic reviews aim to provide a comprehensive overview of empirical evidence that fits pre-specified eligibility criteria in order to answer a specific research question (Munn et al., 2018). The choice of doing a scoping review and not a systematic review was due to the time limit of the thesis project. The interview study was conducted with qualitative content analysis. Qualitative methods are suitable when the field of study is relatively unknown, and when there is no clear hypothesis made in advance. Methods are further described in chapter 3.

1.8 Delimitations

This thesis project does not provide a complete compilation of WEE methods described in research, as would a systematic literature review. Neither does this project present a representative presentation of how Swedish organisations evaluate economic effects of their work environment, since the sample of organisations is small and made through convenience sampling.

1.9 Structure of the thesis

Chapter 2 presents relevant background information about systematic work environment management (SWEM), WEE and health economics (HE). Chapter 3 presents the methods used in the project. Chapter 4 presents the results of the literature review, and the qualitative analysis of the interviews conducted. In chapter 5, the results and the methods are discussed and suggestions for future research are presented. In chapter 6, the conclusions from the completed work are provided.

2. Background

In this chapter, basic background information about Work environment economics and Health Economics is provided, and the relation between the two is described. Additionally, essential concepts within the field are explained. Previous related work is presented and connected to this thesis.

2.1 Systematic work environment management

Systematic work environment management (SWEM) involves investigating, conducting risk assessments, implementing and following up on the operation in such a way that ill health and accidents are prevented. SWEM can be described as activities in a continuous improvement effort with the aim of managing and preventing work environment risks in the organisation. According to the Swedish Work Environment Law (Arbetsmiljölag (1977:1160)), the employer is responsible for systematically planning, managing and controlling the operations in a way that ensures that the work environment meets the prescribed requirements for a good work environment. However, employee participation is crucial for SWEM to be successful, and this is also stated in the same law: "*Employers and employees shall cooperate to achieve a good working environment.*" (Arbetsmiljölag 1977:1160, ch.3, § 1a). In Sweden, the provision AFS 2001:1 from the Swedish Work Environment Authority (SWEA) prescribes *how* the employer should organise SWEM.

Even though systematic work environment management has long been regulated by law and has support from both employer and employee parties at a central level, SWEM is not implemented in all organisations. The yearly work environment report from SWEA regarding 2024 (Arbetsmiljöverket, 2025) reports that 70 % of Swedish employees know who is responsible for work environment management at their workplace and almost half, 47 %, believe that regular risk assessments are carried out to avoid ill health and accidents at their workplace. This figure has remained relatively stable at around 50 % over the last twenty years (Sjöström et al., 2019). It has been found that the adherence to SWEM differs between industries and between organisations of different sizes. In general, large organisations more often than small ones have a well-developed SWEM and SWEM can be completely absent in many small organisations (Johrén, 2017).

2.2 Work environment economics

The work environment is everything that affects us as humans in the workplace, whereas economics is the management of available resources (Lagerström et al., 2019). Lagerström and colleagues (2019) have defined work environment economics as "the careful management of the human as a resource" (Lagerström et al., 2019). In this thesis, work environment economics (WEE) is defined as the examination of how workplace conditions, structures, and resource constraints influence employee well-being, productivity, and organisational outcomes. It considers the allocation of scarce resources, and the trade-offs involved in optimizing work environment interventions for both efficiency and sustainability (Aboagye & Johanson, 2025).

2.2.1 Tools for relating work environment to business results

The literature review by Rose et al. (2013), provides an overview as well as an evaluation of nine models used to connect WE to business results. As can be seen in Table 1, the study by

Rose et al. (2013) divided the different tools into three different categories: 1) Analysis of WE risks and calculation of their economic impact, 2) Development and evaluation of suggestions for intervening measures in combination with investment analysis and 3) Investment analysis in connection with work environment (WE) investments.

Table 1. Work Environment Economic Tools described, evaluated and categorized in Rose et al. (2013).

Category 1: Analysis of WE risks and calculation of their economic impact	**Category 2: Development and evaluation of suggestions for intervening measures in combination with investment analysis**	**Category 3: Investment analysis in connection with WE investments**
The Tool Kit (Amador-Rodenzo, 2005)	ROHSEI (Linhard, 2005)	The Net-Cost Model (Lahiri et al., 2005)
SCA (Rose & Örtengren, 2000)	The Balloon Model (Johanson & Johrén , 2001)	The Potential (Bergström, 2005)
MAWRIC (Rose & Örtengren, 2000)		The ProductAbility Tool (Oxenburgh et al., 2004)
WEST (Karling Brohammer, 2002)		

2.3 Health economics

While WEE concerns the evaluation of economic effects of risks, costs and interventions *in the workplace,* health economics most often analyses how resources in healthcare and public health are used to improve people's health in an effective way. Health economic evaluations (HEE) may be defined as" the *comparative analysis of alternative courses of action in terms of both their costs and consequences*" (Karolinska Institutet, n.d.) and is used to provide guidance for decisions aiming at improving the efficiency of the allocation of resources to improve health. HEE methods are widely used for evaluating healthcare but can also be used for comparing worksite health promotion alternatives (Lutz et al., 2020).

2.3.1 The most common HEE methods

The most common HEE methods are cost analysis (CA), cost-effectiveness analysis (CEA), cost- utility analysis (CUA) and cost-benefit analysis (CBA) (Umeå University, n.d.). According to Grimani et al. (2018), only the results of such full economic evaluations should be used if one wants to draw conclusions about the cost-effectiveness of OSH. Full economic evaluation means that models are *simultaneously* analysing costs *and* health effects of an intervention in the context of worksite health promotion *compared with* one or more alternatives. The alternative can be another intervention, or no intervention (Drummond, 2015). In CEA, CUA and CBA, *costs* are always expressed in monetary units, while *effect sizes* can be expressed in terms of natural units (CEA), monetary units (CBA) or quality-of-life proxies (CUA) (Lutz et al., 2020). Examples of natural units are for example the number of lives saved, or the number of days at work during a year. Examples of quality-of-life proxies are Disability-Adjusted Life Year (DALY), a measure of lost healthy life years, and quality-

adjusted life-year (QALY), a measure of how many years of life are lived in good health (Osmosis by Elsevier, n.d.).

2.3.2 Quality of economic evaluation studies

It has been stated that economic evaluation studies of OSH would benefit from more methodological standardization (Steel et al., 2018). Moreover, guidance and recommendations concerning the measurement and valuation of productivity-related costs as well as the statistical analysis of economic evaluations have been requested, in order to improve the validity of results and conclusions of economic evaluations in occupational health (van Dongen & van der Beek, 2022). The methodological quality of health economic interventions evaluations in healthcare, especially within systematic reviews, can be assessed by using the Consensus Health Economic Criteria (CHEC) list (Evers et al., 2005). The checklist includes 19 criteria, covering aspects such as perspective of the analysis, identification of relevant costs and outcomes, description of study population and alternatives, measurement, valuation of costs and outcomes, time horizon, discounting and sensitivity analysis (Evers et al., 2005).

2.4 The relationship between work environment economics and health economics

WEE has its roots in personell economics, a sub discipline of business economics, and uses calculations for analysing how a better WE can affect the economy both for the organisation and the society. HE is a subdiscipline of national economics and uses regression analysis to study relationships between resources, healthcare quality and health outcomes with the aim of maximizing public health for the money spent. WEE can be viewed as a hybrid between personnel economics and health economics because it draws on theories, methods, and goals from both fields to analyse and improve the economic aspects of workplace conditions.

2.5 Financial effects of the work environment

The work environment has been shown to have various financial implications for organisations. The most significant cost factors are outlined in the following section.

2.5.1 Hidden and visible costs

A challenge when making WEE calculations is that some financial consequences of the WE are accounted for, while others are not (Johansson & Johrén, 2017). Sick leave is an example of a “visible” consequence, that is directly seen in the external accountancy, while other consequences of a poor work environment are “hidden” and are not explicitly accounted for, e.g. lower productivity, quality or lower sales. As a result, the full economic consequences of poor working environments are generally unknown to decision makers, and organisations often lack insight into the actual costs associated with poor working conditions as well as the potential benefits of targeted interventions (Abrahamsson, 2020; Rose et al., 2013). So, what consequences should be included in a WEE calculation? As Johansson (2017) states; if the organisation’s steering principles are

based on accounted costs, a calculation that shows improvements in productivity will not be impactful. At the same time, when including hidden costs in a calculation, the financial consequences might be so large, that it undermines the confidence in the calculation (Johansson & Johrén, 2017).

2.5.2 Terminology

Some authors in the WEE field have used the terminology “Direct” and “In-direct costs” instead of Visible and Hidden costs. This can cause confusion, since the terms Direct and In-direct costs are traditionally used with another meaning within accountancy:

- *Direct costs* - costs that can be clearly and directly attributed to a specific product, service, project, or department, e.g. raw materials in manufacturing, salaries (Johanson & Johrén, 2017).

- *In-direct costs* - costs that cannot be directly traced to one specific product or person — they support multiple activities and are spread across the organisation, e.g. Overhead costs, IT infrastructure.

The In-direct costs are shared costs, not tied to one product or project. To accurately determine the production costs of a specific product or service, it is necessary to allocate indirect costs using an allocation model (Johansson & Johrén, 2017).

Already in 1931, Heinrich introduced the idea of visible and hidden work environment costs which has been thereafter illustrated in different Iceberg Models, for example by International Labour Organization (ILO), see Figure 2. In this thesis, his terms *Visible* and *Hidden* costs are used when discussing financial effects of the work environment.

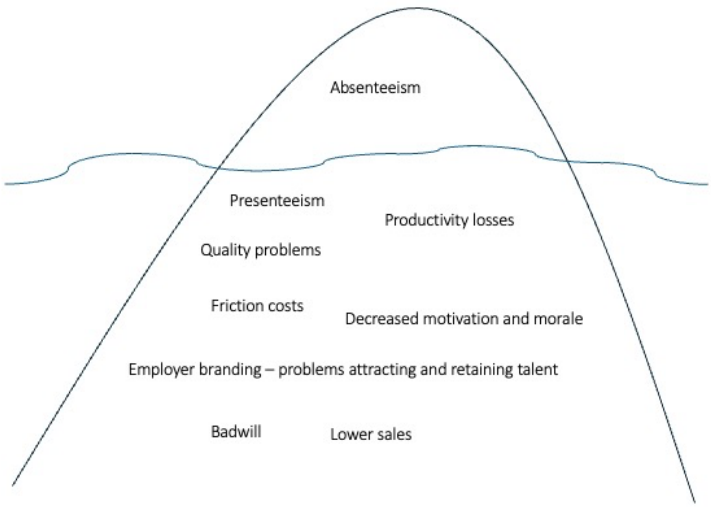


Figure 2. *The Iceberg model of visible vs hidden costs for work environment problems (Modified after International Labour Organization, 2007, p. 4).*

2.5.3 Productivity

Productivity may be defined as the ratio between results achieved and resources invested (Lagerström et al., 2019). Productivity loss on the other hand, has been defined as a measure of the reduction in employees' performance (Johansson, 2017). In other words, according to economic theory, productivity is assessed as output, which is a function of capital and labor input. Productivity loss is consequently associated with reduced labour input, as a result of absenteeism and presenteeism (Strömberg et al., 2017).

Deficiencies in productivity and quality are seen as the most prominent costs of a poor work environment among most researchers. Nevertheless, there are few high-quality studies, such as randomised controlled trials (RCT's) that have investigated these connections (Johansson & Johrén, 2017). How productivity is measured differs among studies. Some researchers use self-rated productivity, whereas others use the substitute concepts absenteeism and presenteeism for measuring productivity (Johanson & Johrén, 2017; Lohela-Karlsson, 2010).

Work environment–related productivity loss was defined by Strömberg et al., (2017) as any physical, psychological, or social problems that might arise in the work environment and impair work performance. The authors found that job-specific characteristics, such as the amount of teamwork, the ease of substitution of workers, and time sensitivity of output, can implicitly affect how much productivity loss incurred by an organisation (Strömberg et al., 2017). Lohela Karlsson et al. (2013) found that poor work environment seemed to be more costly for organisations than health problems, when only considering reasons related to reduced performance while at work (Lohela Karlsson et al., 2013).

2.5.4 Absenteeism

Absenteeism is the time an employee is not present at work due to illness or other health-related issues (Goetzel et al., 2004). According to a report from Skandia Insurance (Skandia, 2024) the societal cost for missed workdays due to long term sick-leave in Sweden during 2024 amounted to SEK 92 billion. The sick-leave costs for women amounted to 54,2 billion, and for men 37,7 billion (Skandia, 2024).

The organisational cost of sick leave is relatively easy to calculate, compared to estimates of productivity and quality. There are also templates for sick leave that can be used in financial calculations (Johanson & Johrén, 2017). Johrén (2017) has argued that there is a direct link between high numbers of short-term sick leave among employees and a bad work environment in an organisation (Johrén, 2017). However, there are some crucial problems with using sick leave rates when accounting for work environment economic effects. Firstly, it might be difficult to be sure of the connection between the sick-leave and the work environment (Sjöström et al., 2019). The employee has no duty to report the reason for sick-leave, and the employer has no legal right to access information about the employees' health conditions. From the Skandia Insurance report previously mentioned, we do not know how much of the sick leave is caused by work-related ill-health. Secondly, many factors beside the work environment affect the rates of sick leave; e.g. the design of the health insurance and type of work, including possibilities to work from home and even the possibilities for presenteeism, i.e. going to work despite not feeling well (Johrén, 2017). Thirdly, most researchers consider declines in productivity and quality to be more significant costs of a poor work environment than

absenteeism. Sickness absence is often both preceded and followed by periods of presenteeism, during which employees are at work but less productive due to illness (Caverley et al., 2007). As a result, focusing solely on the costs incurred during the absence itself leads to an underestimation of the total economic impact.

2.5.5 Presenteeism

Presenteeism refers to attending work despite illness (i.e., acute or chronic) which would normally justify absence. Previous research has shown that the costs associated with presenteeism—or reduced productivity—often exceed those of sickness absence (Collins et al., 2005) with some studies suggesting they may be two to three times higher (Goetzel et al., 2004; Stewart et al., 2003). Presenteeism can result in lower productivity and quality of the work performed – a hidden cost for the organisation.

The Swedish Social Insurance Agency (Försäkringskassan) received approximately 69,000 notifications about accidents at work after which employees did *not* take sick leave during 2022, corresponding to 13.7 notifications per 1,000 employed (Arbetsmiljöverket, 2023). It is reasonable to assume that at least some of these accidents and lost days at work can be prevented by improving physical, cognitive social and environmental factors in the work environment conditions (Rose et al., 2013).

2.5.6 Friction costs

Friction costs refer to the indirect costs that arise from employee absenteeism or turnover (Koopmanschap et al., 1995). Johansson & Johrén (2017) provide a compilation of friction costs. They show one visible friction cost, advertising for a new employee, whereas many costs are hidden:

- Time spent by management on recruitment
- Production loss before a substitute or new employee is in place
- Production loss during introduction of substitute or new employee
- Introduction courses
- Production loss for mentor to new employee
- Reduced productivity during the notice period due to a natural phase-out in work tasks and declining motivation

Introduction costs vary between different professions, positions and individuals. Johansson and Johrén (2017) estimated that the cost of replacing a trained engineer with a MSc degree amounts to nearly 1.5 million SEK.

2.5.7 Quality

The link between a good physical and organisational work environment (WE) and high-quality output has been described in several studies (Abrahamsson, 2000; Nembhard & Edmondson, 2006). Kolus et al. (2018) systematically reviewed 73 empirical studies, studying the linkages between the human-system fit in the design of operations systems (OS) with production quality. Quality risk factors included human factors aspects in product design, process design and workstation design. Quality deficits were associated with high workload like fatigue and injury-related risk factors. Forty-six percent of the studies reported on efforts to improve HF in the OS with effect sizes for quality improvements reaching up to 86% (Kolus et al., 2018).

Abrahamsson (2000) evaluated the effects of improved working conditions in the steel industry during the relocation of operations from an old facility, where both sickness absence and presenteeism had been prevalent, to a new facility. The enhanced working conditions at the new site led to improvements in both production quality and efficiency. However, Abrahamsson also highlighted several challenges in conducting financial follow-ups on investments in the work environment. While personnel statistics were readily available, tracking production-related outcomes proved difficult due to the company's overly simplified and non-specific follow-up system. Additionally, Abrahamsson (2000) noted the complexities involved in determining which changes could be attributed to the new workplace and in assigning them a monetary value. She systematised the economic effects of the investment by categorising them into four groups, see Figure 3.

Despite the described challenges, the profitability analysis indicated that investments aimed at improving the work environment can result in strong economic returns. Of large interest is that the analysis showed that only 2% of the financial benefits were related to sick-leave cost savings, whereas 98 % were related to productivity and quality improvements.

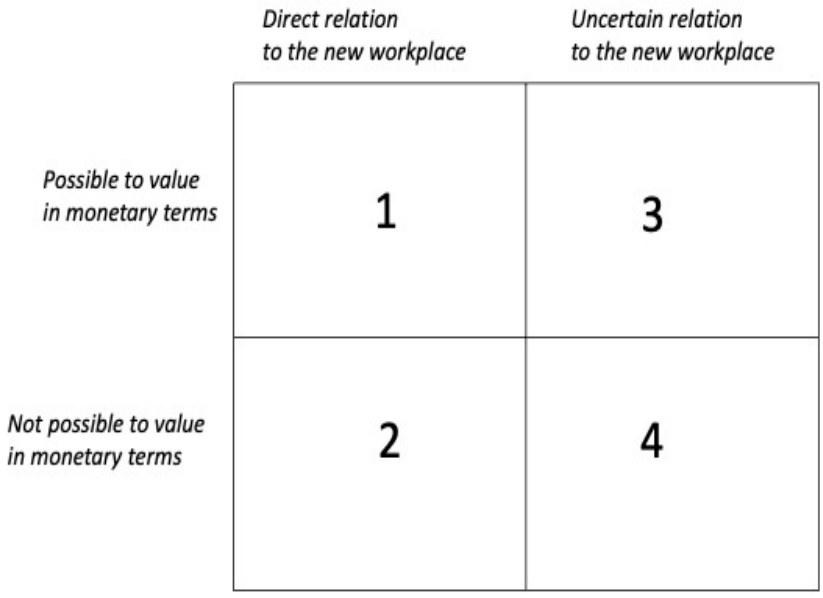


Fig. 3. *Four different categories of economic effects, according to Abrahamsson (2000, p. 4).*

2.5.8 Accounting for investment in employees' health and well-being

Just as in Abrahamsson's case, challenges are often met with regards to traditional accounting procedures when accounting for financial effects of the work environment. One potential problem is that employees are not seen as an asset in the organisation's balance sheet, and investment in the employees' health and well-being can therefore not be depreciated over several years, as can for example investments in ergonomic equipment or other inventory for improving the work environment. Investments in employees, such as training, need to be accounted as a cost in the organisation's external accounting, that immediately affects the organisation's result. Consequently, investments in employees might meet a resistance since short-term results are often requested by owners and other stakeholders. This is particularly challenging because the effects of training often only become profitable over a longer time horizon, while the effects cannot always be traced back to the investment as clearly as, for example, when purchasing new equipment (Lagerström et al., 2019).

2.6 Key performance indicators

Key Performance Indicators (KPI's) can be used both for external communication and benchmarking, and for internal communication for goal setting, follow-up and evaluation (Lagerström et al., 2019). KPI's are expressed in numbers as monetary and non-monetary measures. Non-monetary KPI's are usually easier to interpret, while monetary KPI's can be advantageous in financial decision making (Johanson & Johrén, 2017). Non-monetary KPI's are for example yearly *days* of sick leave, while a monetary KPI is yearly *costs* of sick leave. It is also possible to calculate an KPI of effect, where an achieved effect, or revenue, is compared to the resource used to create the revenue. KPI's can be proactive (steering) or reactive (lagging). Whereas reactive KPI's show the results of activities done and are often easier to measure in for example costs, proactive KPI's can be more difficult to measure, but more efficient in steering a behaviour which can be of big importance in SWEM. It has been stated that work environment management traditionally has a reactive focus (Berlin & Adams, 2017), e.g. using lagging KPI's for reported incidents and accidents or lost time injuries. For a proactive and systematic work environment management, instead of focusing only at risks, a change of focus to the desired, safe behaviours are helpful (Matton & Rosengren, 2018; OSHA, 2019). Leading KPI's formulated in terms of desired OSH goals can then function as motivators. Pawlowska (2015) conducted a study in 60 companies in order to determine what kinds of indicators were used for OSH performance measurement by companies with different levels of OSH performance. The findings revealed that all companies primarily used indicators focused on meeting statutory requirements. However, companies with higher OSH performance levels were significantly more likely to adopt leading indicators (Pawłowska, 2015).

2.7 Ethical aspects of WEE

Is it possible to put a price on a work-related injury or a healthy worker? And is it fair? There is a risk that WEE calculations can be used for more efficient management of personnel as a production resource, which would be incompatible with the intrinsic value of humans (Johanson & Johrén, 2017). The idea of intrinsic value of humans is that every human is irreplaceable, and with such a stance, it would be wrong to account the value for a human as a production resource. Johansson & Johrén (2017) argue, that when making a cost/benefit analysis, a price is not placed on the person but on different consequences of measures that involve humans. The goal of WEE analyses should be to demonstrate mismanagement, for the benefit of human value. To ensure this, WEE needs to be combined with a strong personnel policy (Johanson & Johrén, 2017).

2.8 Related work: A survey and evaluation of methods for calculating the financial impact of the work environment

As earlier mentioned in the Introduction chapter, Rose & Orrenius (2007) compiled information on methods and models for calculating the financial impacts of the work environment. Specifically, their objective was to describe and evaluate existing methods based on questions related to methodology, target audience, documentation, and applications. Group interviews were conducted with representatives from 17 Swedish organisations within different business sectors. The authors found that only a few of the participating organisations used methods to link work environment and economy, and none of the representatives contacted believed that the methods they had access to were satisfactory. Rose & Orrenius (2007) pointed out suggested reasons for not using WEE methods, where one was that existing methods might not be successful in linking work environment factors to the organisation's core values. Another reason for not using WEE methods was lack of knowledge about existing models. Other potential reasons were that interviewees did not consider themselves to have such needs, since work environment improvements measures might be implemented without economic arguments, or that the methods were judged to be difficult to understand and use. The interdisciplinary approach needed for effective utilisation of these methods was also seen as a potential difficulty (Rose & Orrenius, 2007).

The following is a selection of recommendations for further development of WEE methods presented by Rose & Orrenius (2007):

1. Adaptation to current legislation and concepts and language commonly used by the users are advantageous
2. Simple methods using standard values are easy to apply, but their disadvantage lies in potentially low accuracy. Methods that more comprehensively account for factors influenced by the work environment offer more accurate estimates of the true effects.
3. Methods should be transparent, which creates trust and acceptance within the organisation.
4. Hidden financial effects, such as quality, psychosocial factors, motivation, good-will and productivity should be included in the model. Only considering visible costs, e.g. for sick leave, is insufficient.

3. Methods

The purpose of this chapter is to provide an overview of the research methods used in this thesis. Section 3.1 describes the process of the literature review. Section 3.2 details the qualitative methodology of the interview study and approaches used for the data collection and data analysis in this research.

3.1 Literature review

The first aim of the scoping literature review (Munn et al., 2018) was to identify and provide a description of the types of WEE methods described in the field of WEE research during 2012-2024. The methods should be related to the organisational financial impact of work environment risks, problems and interventions. The second aim was to identify and analyse knowledge gaps between methods used in research and methods used in practice by the organisations participating in the interview study.

3.1.1 Literature search process

After initial test searches, the final literature searches were performed with assistance of a librarian from the KTH library, in January 2025. The databases used were Web of Science, PubMed and Ergonomics Abstracts. Searches were performed using **key words,** including synonyms and MesH terms, originated from the search query "What **models** have been described in research after 2012-2024 on **economic evaluation** of **organisational financial impact** of **work environment**?" Keywords and synonyms are presented in Table 3. Inclusion criteria are presented in Table 2. All types of articles including a description of a WEE method were included. Studies including health economic evaluations, but without an occupational health perspective, were excluded. Articles including health economic evaluations of health conditions for which the connection to the work environment was considered non-specific, such as gastrointestinal disorders or substance use disorders, were also excluded. It was noted whether the authors described a study where the method was used, or if the method was only described per se.

Table 2. *Inclusion criteria.*

	Included	**Excluded**
Language	English and Swedish	All other languages
Publishing years	2012-2024	Studies published before 2012-01-01 or after 2024-12-31
Population	Workers: All professions and type of workplaces	Studies without occupational health focus, e.g. clinical studies of health outcomes for patients with a specified diagnosis
Peer review	Peer reviewed articles, conference proceedings as well as grey literature	
Type of article	Articles including descriptions of economic evaluation of costs, risks or investments related to work environment from a business (financial) perspective, in opposition to societal perspective. Articles with descriptions of economic evaluation methods, without utilisation of the method in a research project.	Articles including evaluation of health outcomes of occupational health interventions without economic evaluation. Health economic evaluation of organisational costs, risks or investments in relation to diseases without clear connection to the work environment, e.g. obesitas or substance use disorders.

Table 3. *Keywords and synonyms.*

Number	Keyword	Synonyms
#1	Model*	Method* Tool* "Economic evaluation*" "business result*" "bottom line" "Return on investment" "cost analysis" "cost effectiveness" "cost-efficacy" "cost-benefit" "cost benefit analysis" "investment analysis"
#2	"economic effect*"	"Economic impact" "financial effect*" "financial impact*" Cost Productivity Quality
#3	N/A *	"personnel turnover" "Sick leave" sick-leave presenteeism absenteeism goodwill badwill
#4	Organisation*	Organisation* Compan* Business*
#5	"Work environment*"	"Working condition*" "Work-related"

* No overarching term was found for these keywords.

Table 4 presents the search logic from start until finish. To get a high sensitivity in the search process, an initial wide search was conducted in Web of Science. This was done to retrieve all relevant articles within the research area, or at least discover as many as possible (Karolinska Institute Library, n.d.) A consequence of this search process was that not all search results were relevant. In systematic literature searching, a precision of two to three % is common (i.e., 2-3 % of the hits are relevant). Since the scope of this thesis is smaller than that of a

systematic literature review, the search needed to be narrowed. In contrast to a wide search, a narrow (or *precise*) search results in a larger share of relevant hits but also implies the risk of missing relevant studies (Karolinska Institute Library, n.d.). In this case, with a limited amount of time for the review, that risk was considered acceptable. By structuring the search terms in blocks with Boolean logic (MIT Libraries, 2021), the search was refined and resulted in a manageable number of relevant hits. Boolean operators form the basis of mathematical sets and database logic (MIT Libraries, 2021). In this search, the Boolean operators **AND** and **OR** together with proximity operators, were used to narrow the set of results. By using proximity operators, a certain proximity is specified between the search terms. Proximity operators are more specific than Boolean operators and make the search more precise (MIT Libraries, 2021).

Table 4. *Search logic (from start 2025-01-23 until finish 2025-01-24).*

Search no.	Database	Search terms	Notes	No of Hits
1	Web of Science	#1 AND #2 AND #3 AND #4 AND #5	Restriction: Topic Proximity operator: max 5 words between #2 and #3	325
2	Ergonomics Abstracts	#1 AND #2 AND #3 AND #4 AND #5		49
3	PubMed	#1 AND #2 AND #3 AND #4 AND #5	Restriction: Title/Abstract	95

In total, 17 original articles and five systematic reviews were included. The articles in the systematic reviews were not included in the current thesis due to the project's limitations and the publication year criteria. The inclusion process can be followed in Table 5. The economic methods identified were described but not evaluated, as such evaluation fell outside the scope of this thesis project. The result of the literature search was actively compared to the result of the literature search in Rose et al. (2013), and the included articles were divided into the same three categories as was presented in Rose et al. (2013), with an adaptation made for category 1: 1) Identification of WE risks and calculation of their economic impact, 2) Development and evaluation of suggestions for intervening measures in combination with investment analysis and 3) Investment analysis in connection with work environment investments. After the database searches were completed, a complementary survey of grey literature was performed and an additional fourth category was added for those results.

Table 5. *Inclusion process.*

Total number of articles after finished database searches	469
Duplicates	68
Excluded articles according to the inclusion criteria, after reading abstract	341
Excluded articles according to the criteria, after reading whole article	38
Included articles	22
Whereof - Systematic reviews	5
- Original articles	17

3.2 Interview study

The interview study followed the seven steps of an interview inquiry as described by Kvale (2007): Thematizing, designing, interviewing, transcribing, analyzing, verifying and reporting.

3.2.1 Thematising and designing

The method chosen for the interview study was qualitative content analysis as described by Graneheim & Lundman (2017) and Kvale (2007). The choice of method was discussed with a researcher with extensive experience in interview studies and qualitative analysis. Qualitative content analysis was chosen since it is considered well suited for structured interviews with relatively specific research questions, and when the aim of the analysis is to analyse the manifest content of the data (Kvale, 2007). *Manifest content* is defined as visible, surface-level content, i.e., what is explicitly said, whereas *latent content* refers to the underlying meaning behind the words — what is implied or inferred (Kvale, 2007).

3.2.2 Interviewing

3.2.2.1 Interviewees

Interviewees were recruited via email through convenience selection via personal and professional networks of the author and the supervisor. In total, individuals from nine organisations were invited to participate in the study. Individuals from five organisations volunteered to participate. Each included organisation was represented by one to three interviewees. In total, eight interviews were conducted, seven with one interviewee and one

with two interviewees. See Table 6. Among the interviewees, six were men and three were women. The invited organisations that declined to participate in the study represented the electronics (1), pharmaceutical (1), and healthcare (2) sectors.

Table 6. *Included organisations and interviewees.*

Organisation	No of interviewees	Profession(s) interviewed
Manufacturing company A	2	Ergonomist, OSH manager
University	1	OSH specialist
Manufacturing company B	3	Ergonomist, safety delegate, work environment coordinator
Hospital	1	OSH manager
Logistics company	2	OSH specialist (2)

3.2.2.2 Interview guide

An interview guide was developed grounded on the research questions and with supervision from an experienced researcher within the field of qualitative interview studies. A pilot interview was conducted with a student colleague with long experience in systematic work environment management, and the guide was thereafter adjusted to some extent. The interview guide can be found in Appendix D.

3.2.2.3 Interviews

The interviews were conducted during March and April 2025. Two interviews took place at the interviewee's respective workplaces and six via Zoom. The duration of the interviews was 40-70 minutes (M= 55 minutes). Before the interview, participants were given information about the study in written and oral form and signed their informed consent according to KTH's research procedure. The interviews were recorded using Zoom or Voice Memo on an iPhone 13. Sound files were then uploaded to the author's private Microsoft OneDrive.

3.2.3 Transcription

The Swedish Royal Library's speech-to-text AI model KB-Whisper (Library, 2025) was used to transcribe the sound files to text. KB-Whisper processes the text locally on the computer's hard drive. After transcription, the generated text files were exported to Word format and a re-listening of the interview was conducted, correcting all mistakenly transcribed words.

3.2.4 Analysis

For the qualitative content analysis, an inductive, bottom-up approach was used with the research questions as a starting point, as described by Babapour Chafi & Rolfö (2019). This involved iterative reading of the transcripts, coding of meaning units, identification of recurring categories and overarching themes, and linking the themes to each other in a final analysis (see Appendix E).

3.2.5 Verification

The verification process started at the end of each interview, by the interviewer summarising the most prominent themes and asking the interviewee about his or her thoughts on the suggested themes (*member validation*, according to Kvale, 2007). The analysis was also *peer-validated* (Kvale, 2007) by a researcher familiar with the interview themes.

3.2.6 Reporting

The identified themes were *i)* Perceived key cost factors associated with poor OSH, *ii)* Known and/or utilised WEE tools, *iii)* Drivers for utilising WEE tools, *iv*) Motives for not utilising WEE tools, *v*) Alternative approaches of addressing WEE effects, and *vi)* Ideas for (further) increasing organisational awareness on WEE.

4. Results

In this chapter, the results from the literature review and the interview study are presented.

4.1 Results of the literature review

This section presents the economic methods identified in the scoping review, divided into the same categories as presented by Rose et al. (2013), with an adaptation made for category 1. Category 4) presents findings from grey literature.

4.1.1 Findings in category 1: Identification of WE risks and calculation of their economic impact

Unlike in the review by Rose et al. (2013), no articles were found that combined a risk assessment or risk analysis of the work environment with calculation of the economic impact. The seven articles included in category 1 were instead used to calculate the costs of an identified risk in the work environment, e.g. work-related stress or unspecified work environment problems. In total, six different methods were used in the seven articles. The following section presents five of the methods, while formulas can be found in the List of Economic Methods. The sixth method (Cho et al., 2023) is not described in further detail due to its complexity and the consideration of limited usability for OHS professionals. A complete list of the six methods can be found in Table A1.

Table 7. *Five economic methods found in category 1: Calculation of the financial impact of WE risks.*

	Method	**Article(s) describe the method as being used in research and/or practice**
1.	The human capital method (HCM)	Yes
2.	Friction Cost Method (FCM)	Yes
3.	Systematic Accident Cost Analysis Method (SACA)	Yes
4.	Assessment of the relation between productivity loss costs and presenteeism	Yes
5.	Wage multipliers for estimating the costs of productivity loss	No

4.1.1.1 The Human Capital Method (HCM)

The Human Capital Method (HCM) is a commonly used approach in health economics and cost-of-illness studies to estimate the economic impact of disease or injury in terms of lost

productivity (Pike & Grosse, 2018). The method is primarily used for estimating economic impact on the societal level but can also be used on the organisational level. The value of lost productivity due to sick leave, disability or premature death can be estimated. This method values all time lost due to illness, disability, or premature death as a loss of productivity over the entire period of absence (e.g., until retirement or death).

4.1.1.2 Friction Cost Method (FCM)

The Friction Cost Method (FCM) also estimates productivity losses due to illness, disability, or premature death—but in a more realistic and conservative way compared to HCM (Pike & Grosse, 2018). Unlike HCM, this method values only the productivity lost during the *friction period,* i.e. the time it takes to replace the sick worker. Just as HCM, this method is primarily used on the societal level, however, it can be applied also on the organisational level. In practice, organisations often rely on simpler direct cost estimations (e.g., wages, sick leave costs) rather than a full FCM framework (Koopmanschap et al., 1995).

4.1.1.3 Comparison between HCM and FCM

Pike & Grosse (2018) made a systematic review of 80 studies from Germany, The Netherlands and Canada using the FCM for calculating the Cost of Illness, with or without comparison to the HCM. They stated that HCM was the most used method for estimating productivity costs, although it has been criticized for potentially overestimating productivity losses. In contrast, FCM—which focuses on the short-term costs of replacing absent workers—was used in only about 5–8% of cost-of-illness studies. Their results point at that estimates of the economic impact of chronic conditions tended to be significantly lower when the friction cost method was applied, although the authors stated that the results can vary depending on how each approach is implemented. A lack of standardisation between methods limits the comparability of productivity cost estimates (Pike & Grosse, 2018).

4.1.1.4 Systematic Accident Cost Analysis Method (SACA)

The Systematic Accident Cost Analysis Method (SACA) method concentrates on the consequences of occupational accidents and the associated costs, categorised into six broad groups (Popescu-Stelea et al., 2019). Its primary process involves first identifying the activities triggered by the work-related injury under analysis, followed by calculating the costs these activities incur for the organsiation. The method relies on a set of forms and checklists intended solely as guidance. Its main objective is to equip managers with a practical tool for identifying accident-related costs—without requiring extensive expertise in financial analysis. Based on interviews, the SACA method is not designed for real-time application, but for more generally gaining knowledge and raising awareness of accident-related costs within the organisation (Popescu-Stelea et al., 2019).

4.1.1.5 Assessment of the relation between productivity loss costs and presenteeism

Vänni et al. (2017) have developed a method for calculating the total loss in days due to presenteeism at a company in food industry (factories and administration). The method is

based on employee self-ratings on The Productivity Loss Scale and the company's sickness absence register data (Vänni et al., 2017). The full formula can be found in Vänni et al., (2017), p. 36.

4.1.1.6 Wage multipliers for estimating the costs of productivity loss

Strömberg et al. (2017) derived wage multipliers that can be used by managers to estimate the costs of productivity loss for employers in economic evaluations.

The authors modelled a prediction of the extent of productivity loss of absenteeism resulting from job characteristics after gathering data from a survey panel of 758 managers from different business sectors. The predicted values were then used to derive wage multipliers based on the cost of productivity estimates provided by the managers, see Table A1.

4.1.2 Findings in category 2: Development and evaluation of suggestions for intervening measures in combination with investment analysis

In category 2, one study was found that combined a risk analysis with suggestions for intervening measures and a following investment analysis (Ramos et al., 2015). A risk analysis was made of the work accidents in a hospital. Three of the most common types of accidents based on statistics from the previous year were selected: Needle stings, falls and excessive strain. Following the risk assessment, some preventive measures were designed, and the Benefit/Cost ratio (B/C) of these measures were calculated, both in financial terms (from the organisation's perspective) and in economic terms (including the benefits for the worker and for the Society). The benefits exceeded the costs only in some cases. When the societal perspective was considered, the B/C ratio increased significantly (Ramos et al., 2015).

4.1.3 Findings in category 3: Investment analysis in connection with work environment investments

In this category, nine original articles and four systematic reviews were found.

4.1.3.1 Findings from original articles

In the original articles describing investment analysis in connection with work environment invetments, only *organisational* interventions, aiming at promoting better physical and/or mental health amongst employees or preventing ill- health, were described, in contrast to investments in terms of *technical* solutions or *engineering controls* - no such articles were found. The organisational interventions, of which a detailed summary can be found in Table A3, included e.g. screening for burnout and depression, physical activity during working hours and behavioural interventions aiming for reducing sedentary time during the workday. Table 8 provides an overview of identified economic methods described in the current study regarding WE investments.

Table 8. *Economic methods used in the nine original studies included in Category 3. For details about the included articles, please see Table A3.*

	Method	**Used in research and/or practice**
1.	Cost-Effectiveness Analysis -Sub group: Within trial Incremental cost-efficacy Analysis	Yes
2.	Cost-Benefit Analysis -Sub group: Only Cost/Benefit Ratio	Yes
3.	Return On Investment Analysis	Yes

4.1.3.2 Findings from systematic reviews

Out of the four systematic reviews included, three of them were performed to evaluate the health economic effects of OSH interventions (Grimani et al., 2018; Jacobs et al., 2017; Pieper et al., 2019), while one (Steel et al, 2018) focused on productivity estimates used in economic evaluation of OSH interventions. Economic evaluation methods described in the reviews are presented in Table 9.

Table 9. *Economic evaluation methods used in the four systematic reviews included in category 3*

Author	Country	Method(s) used to calculate financial impact of WE investments	Comments	Included studies
Grimani et al. (2018)	Europe, Canada, United States, Australia	Cost-Effectiveness Analysis (CEA), Cost-Benefit Analysis (CBA), Cost-Utility Analysis (CUA), Return On Investment Analysis (ROI)	All studies included description of economic evaluations models used	19 RCT:s published 2005 - 2016
Steel et al. (2018)	Mainly Europe and North America	CEA (79 % of included studies) CUA with QUALY as outcome measure (21 % of included studies) Most studies using CUA, also used CEA.	N/A	90
(Pieper et al., 2019)	USA, Canada, Australia, New Zealand, and Europe	ROI	Several studies did not use full economic evaluations but reported absenteeism, productivity, net benefit compensated healthcare costs, other healthcare costs and intervention costs	74 reviews published 2012 – 2017, including 1715 single studies
(Jacobs et al., 2017)	Canada	CEA (1 study) Cost/benefit ratio (2 studies)	6 studies calculated costs or net benefit but did not perform complete economic evaluation.	8

4.1.4 Findings in Category 4: Grey literature

In addition to the methods found in database searches, three methods were found in the grey literature and are described below.

4.1.4.1 Occupational Health Economic Analysis Tools from Mynak

The Swedish Agency for Work Environment Expertise (Myndigheten för Arbetsmiljökunskap, [Mynak]) has published three occupational health economic analysis tools, adapted for health examinations and life-style habits, mental ill-health and low back pain respectively (Mynak, 2019). The tools are primarily based on ROI (Brigham & Ehrhardt, 2016) but also include the Payback method, the Annuity method and mentions the Present Value Method and Final Value Method (ibid).

4.1.4.2 The Balloon model

The Baloon Model developed by Johansson & Johrén was also described in the review by Rose et al. (2013). The model which is an adaptation of a Cost-benefit analysis, is performed in eight steps and is described in detail in Johansson & Johrén (2017). The company MedHelp also has a tool developed from the balloon model available on their webpage (MedHelp, n.d.).

4.1.4.3 The Flow Module tool

The company Flow Module (Flow Module, n.d.) provides two tools on their web site: One for calculating the total organisational costs for absenteeism and presenteeism, and one for the costs caused by stress-related ill-health. The stress-related tool is based on employee's self-rated stress levels with KEDS-9, Karolinska Exhaustion Disorder Scale. A reference mentioned as ground for the calculation model is Lohela Karlsson et al. (2013).

4.2 Results of the interview study

In the following section, the results from the content analysis of the eight interviews are presented.

4.2.1 Perceived key cost factors associated with poor OSH

The interviewees described their motives for improving the work environment in connection to their mental models of the key cost factors associated with poor OSH within their organisations, which are shown in Figure 4. Some interviewees could present documented costs, whereas other estimated the perceived largest economic effects of poor working conditions. Different key cost factors were described, depending on business sector. In manufacturing, quality deficiencies were highlighted, while in health care and higher education, the costs of poor organisational and social work environment were emphasised.

"I truly believe that the organisational and social work environment (OSA) and its underlying factors are extremely important from a cost calculation perspective. We know how much even a single conflict can cost in terms of reduced productivity — yet it remains an invisible factor. Conflicts often arise from one of the OSA-related issues, like poor communication or unclear roles."

OSH manager, health care

Presenteeism and long-term sick leave were the most common factors mentioned. The importance of variation in work — not only from an ergonomic perspective but also in terms of intellectual stimulation, was emphasized. This was further linked to employer branding and the ability to attract and retain talent, particularly among younger employees.

”Most young people today want to keep learning. They don’t want to stand on the same assembly line for fifteen years — they want to develop, to a much greater extent than many older employees. Variation is important, but so is the opportunity to learn new things. That’s also a key part of the work environment — feeling intellectually stimulated and having variation in your tasks. It’s a real challenge in an operation like this. (…) And from an employer branding perspective, as well as in terms of retention, it’s an extremely important factor. If you want people to stay, you need to offer them variation.”

OSH manager, manufacturing company A

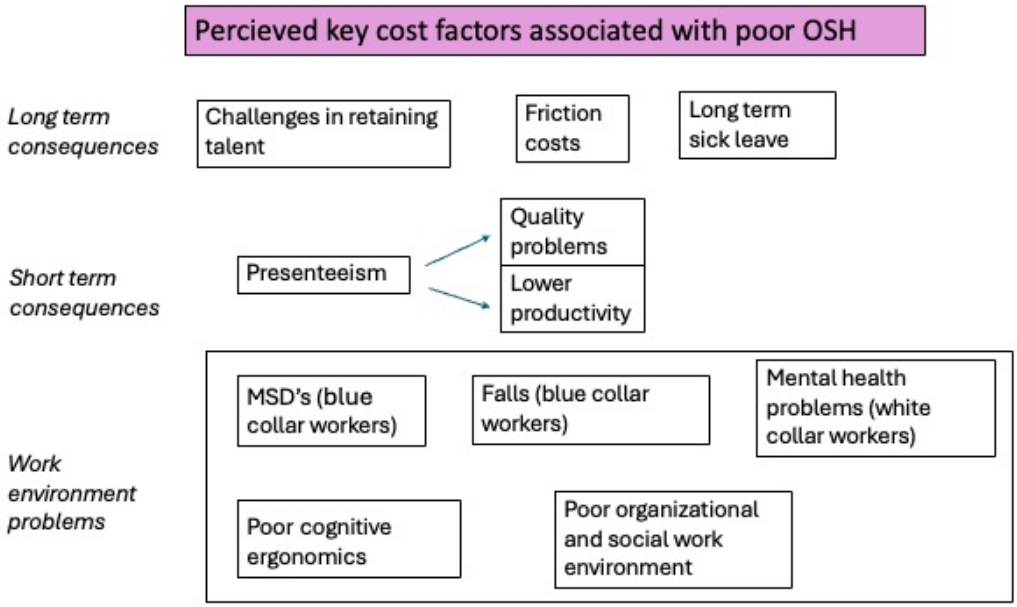


Figure 4. *Estimated or documented key cost factors associated with poor OSH, reported by the interviewees.*

4.2.2 WEE tools – knowledge and use

All interviewees had knowledge of one or more WEE tools, but seven of them reported that no WEE tool was currently being used within their organisation. Two organisations, (within healthcare and higher education respectively), had never used any WEE tools. One organisation used a tool from time to time, previously more often. The other organisations had all previously used WEE tools but had for different reasons stopped. Table 10 shows the WEE tools mentioned in the interviews. Drivers for utilising WEE tools and motives for not utilising them, are further explored in 4.2.3 and 4.2.4.

4.2.2.1 A model for calculation of the costs of poor assembly ergonomics (Falck & Rosenqvist, 2014)

One of the interviewees mentioned the model developed by researchers at Chalmers (Falck & Rosenqvist, 2014) with the aim of enabling calculation of costs of poor assembly quality related to assembly ergonomic conditions. This model was thought of as valid through its fully inclusion of factors that are affected by the work environment, but at the same time complex and time-consuming.

4.2.2.2 Adaptation of The Swedish Social Insurance Agency's Model for Estimating the Costs of Sickness Absence

One of the organisations used an internal model based on The Swedish Social Insurance Agency's model for estimating the costs associated with sickness absence, with some local adaptations made. An explanation to the models' limited use, was its failure to incorporate the economic consequences of productivity losses related to, for instance, presenteeism. In

this organisation, where the knowledge level of SWEM was relatively high, the calculations were not considered as a deal breaker for taking WEE into account:

"The calculation hasn't had much impact on how we conduct our work environment efforts. I believe what has really developed our work environment management is the improvement of our risk assessment methods and the fact that we provide extensive training in how to use them."

OSH professional, manufacturing company B

4.2.3 Drivers for utilising WEE tools

Motivation for performing calculations came most often from the OSH professionals themselves, with the incitement of motivating management for making investments in the work environment on a higher level than is legislated, or even to meet legal requirements, when SWEM was not prioritised within the organisation. WEE methods were seen as beneficial tools for visualizing effects of poor working conditions and raising awareness of the benefits of SWEM.

"We identified a problem area: Employees are not receiving the training in lifting techniques that we had actually agreed upon. People don't take the time to provide this training when new staff members join. So we thought that if we could demonstrate the actual cost of someone developing back, shoulder, or neck problems, it could serve as a motivation to ensure that all new employees receive thorough training in lifting techniques right from the start, as a preventive measure."

OSH professional, logistics company

Motivation for doing WEE evaluations could also come from management. Interviewees from one organisation noted a shift over the past year, with management increasingly requiring OSH professionals to present a business case for proposed investments. They also described emerging tensions between ethical considerations and economic priorities. This shift was attributed to shorter intervals between major projects, which necessitated a higher volume of investments.

In the higher education organisation, WEE models were not used. However, the OSH specialist noted that management often perceived risk assessments and interventions too time-consuming and OSH professionals as overly strict in their internal work environment regulations. The OSH specialist therefore believed WEE tools could be valuable in supporting and motivating SWEM to managers:

"I believe the organisation is expressing a need for this, primarily because they feel we are too governed by rules. Managers say: You also need to account for the administrative costs. How much time am I expected to spend on this for it to truly pay off in terms of improved health and reduced risk? That's a recurring discussion we have. But most of the time, they do accept it. We try to strike a balance and emphasize that these are the minimum requirements."

OSH specialist, higher education

Table 10. *WEE tools mentioned by the interviewees, short description and current usage of the tools.*

Tool	Description of the tool	Number of organisations who mentioned the tool in interview(s)	Number of organisations where the tool was currently in use
Internal calculation method developed by previous colleagues	Calculation for estimating the costs of poor ergonomics	1	0
Falck & Rosenqvist (2014)	The method was meant to be used by engineers and stakeholders in the design or redesign of manual assembly solutions. In addition to costs for work-related sick leave, the module also includes several quality deficiency related costs, e.g. costs for manual assembly errors, costs for recall/repair of products distributed to customers, and costs for lost brand image and customers dissatisfaction (bad will) (Falck & Rosenqvist, 2014).	1	0
Occupational health economic analysis tool, Mynak	See 4.1.3	4	0
Internal calculation method developed by previous graduate	Based on ergonomic risk assessment results for heavy lifting. Estimates costs for sick leave in relation to investment in introductory training.	1	0
Internal calculation method, an adaptation of a tool from The Swedish Social Insurance Agency (Liukkonen)	The method is designed to calculate the fixed, statutory costs during periods of employee sick leave. In addition, overhead costs (OH) may be incorporated into the calculations. The conceptual foundation of the model is partly based on the work of Paula Liukkonen (2004). In the organisations own model, the original model had been completed with friction costs for training a substitute employee	1	1
Flowmodule.se (Rehnberg)	See 4.1.3.	2	0

Another driver that was brought up by an OSH manager in healthcare, who was currently not seeing the need of using any WEE methods, was that calculations could have the potential of reaching out to even more people within the organisation, when arguing for the importance of SWEM:

"But it's probably true that we might reach some people we're currently not reaching — people we don't even realize we're missing. Some are really motivated by numbers and data. But if you look at our Chief Financial Officer, for example — no, it's just not necessary in that case."

OSH manager, healthcare

4.2.4 Motives for not utilizing WEE tools

The motives for not utilizing WEE tools included tool validity, lack of grounding in real life practice and usability.

4.2.4.1 Validity of the tool

Concerns regarding the validity of the tools were raised by several interviewees. Questions about validity came not only from management and financial departments, but also from OSH professionals themselves. One reason for this skepticism was that the tools were considered overly simplified, resulting in low calculation precision. This aligns with the results from Rose et al. (2013). In some cases, the evidence base behind key assumptions was not clearly communicated. Additional concerns included the omission of presenteeism, despite its recognition as one of the largest cost drivers related to work environment problems. Another issue raised was the uncertainty around whether sick leave was work-related or not. Closely related to these concerns were perceptions that WEE tools often lack grounding in real-life practice, as will be described in 4.2.4.2.

"Yes, those were standard estimates, but I don't think people really trusted them. They felt the figures were simply too high. There was a general lack of confidence in the model."

OSH manager, manufacturing company A

4.2.4.2 Lack of grounding in real life practice

Some tools were described as having an epidemiological focus, addressing general health rather than *occupational* safety and health (OSH). As a result, the connection to *work-related* ill-health was perceived as unclear. Another issue raised was that these tools often fail to account for the fact that friction costs (i.e. costs for recruiting or training new workers) vary between different employees.

In healthcare, financial follow-up is complex, and quality is often difficult to quantify in monetary terms:

"Some aspects of quality don't have a clear financial outcome. I mean, if you feel really good after giving birth – how do you account for that? And on the contrary, if you don't feel well afterwards, it may not even be the same clinic that provides follow-up care. The costs are distributed across multiple parts of the system."

OSH specialist, healthcare

4.2.4.3 Usability issues

The time aspect was highlighted by several interviewees. The tools were considered too time-consuming, both in terms of performing the calculations and in becoming familiar with and understanding the research underpinning the models. Also, the OSH professionals own work environment was brought up, with difficulties prioritising when having a lot on one's table and a hard time finding the time for doing calculations.

"It can't be too costly to assess the risk or calculate what sick leave actually costs. Because otherwise, we're way off track. It needs to be simple".

OSH specialist, manufacturing company B

" This kind of occupational health economics mapping you can do, unfortunately, it's an administrative monster."

OSH specialist, higher education

4.2.4.4 Knowledge gap in the organisation

The lack of relevant data within the participating organisations was a factor found to complicate the use of WEE tools. Sick leave data was easily accessible, while data on presenteeism was the most difficult to find across industries. Production metrics were a natural part of daily operations in manufacturing and logistics. In higher education, however, production metrics were more unclear, even though bibliometrics were discussed.

"The statistics we get from occupational health services are practically useless. They can't be applied in any meaningful way. Our sick leave levels are really low, but they don't mean much. They don't reflect reality – we know that. Researchers don't call in sick."

OSH specialist, higher education

Especially in one organisation, the lack of broadly distributed foundational knowledge in ergonomics within the organisation presented challenges when making investment decisions, even when WEE tools were employed. Interviewees described difficulties in demonstrating the return on prevention time to financial departments. To bridge this internal knowledge gap, they suggested educational initiatives in ergonomics as well as opportunities for employees from finance and design departments to gain hands-on experience on the assembly line.

"The problem, to some extent, is when these discussions reach our controllers – they want to see a business case. If you're making an investment, it should pay off in two or three years. But when it comes to the development of work-related injuries or illnesses, that's something we see happening over a much longer period. I find it difficult to respond to that short-term perspective."

OSH specialist, manufacturing company A

4.2.4.5 Other arguments than financial are prioritised

In two organisations, interviewees expressed varying degrees of resistance against calculations regarding ergonomic risks. Their opinion was that legislation, and ethics should be enough motivators for organisations to invest in the work environment, even though there was a demand for financial evaluation from management. In other organisations, the interviewees experienced no need for financial arguments, since the evidence base of WEE was widely spread in the organisation.

"And then, honestly, from an ethical standpoint, it just doesn't feel right. Am I supposed to put a price tag on colleagues in production? Where do we even start that conversation – what's a shoulder worth, what's a back worth? I don't want to go there. (…). If we suspect there's a health issue, we should act on it without being asked to justify it in monetary terms. We already do risk assessments that clearly show the severity of the risks, and we can show how many people are affected – that should be enough. If we also have to put a price on it for someone to take it seriously, then I feel like we're not trusting our own assessments."

OSH specialist, manufacturing company A

"There is general agreement here that investing in safety and ergonomics pays off."

OSH specialist, manufacturing company B

4.2.4.6 Demonstrating the interconnection between WEE and organisational development reduces the need for calculations

Several interviewees highlighted that by recognizing the interconnections between operational development, competence provision and the work environment, the need for economic justification was reduced. A healthcare manager emphasized that a key factor is understanding that beyond the financial burden of sick leave, such absences also lead to a loss of competence on the operational floor, thereby hindering the organisation's ability to achieve its objectives.

"Everything we deal with is about the work environment. If we feel that the job runs more smoothly, that we have more time for the patients instead of struggling with complicated record systems or whatever it may be—then we've made a major improvement in the work environment. So by talking more about the work environment in relation to operational development and goals, we don't even need to talk about money."

OSH manager, healthcare

4.2.5 Proactivity – an alternative approach for taking WEE effects into account

Among the participating organisations, proactivity was found to be the main alternative approach for addressing the economic effects of the work environment. Organisations with this approach relied on trust in the existing evidence on work environment economics, ethics and legislation. This approach was in two organisations used in place of WEE tools, and in two cases, in combination with them. Leading KPI's were part of the proactive approach in some organisations. Organisations that had implemented a structured SWEM approach indicated that it yielded the greatest financial benefits.

Examples from one of the manufacturing companies included:

- Established internal ergonomics standards in the procurement process
- OHS professional involved in design phase of new plants
- Leading KPI's on work environment, e.g. numbers of introductory ergonomics trainings/year

Examples of interventions from the healthcare organisation included:

- Identifying units with a high proportion of staff with frequent short-term absenteeism in order to support their systematic work environment management
- Offering employees access to an in-house physiotherapist within three days when experiencing symtoms of pain or musculoskeletal disorders (MSDs)
- Providing short-term lifestyle counselling for individuals with short-term absenteeism, including a test of aerobic capacity

"What I believe is the single most important thing that has happened for the development of ergonomics is that senior managers and others have come to understand the close connection between quality, productivity, and ergonomics. That insight means you don't always have to rely on detailed calculations. I mean, a production stop of just half an hour or an hour might cost... 100,000. That's a lot of money. Pretty soon, you've paid for a lifting aid. And when you grasp that—maybe not in depth, but at least conceptually—you understand that you must work with ergonomics. The motivation really comes from the fact that it pays off."

OSH specialist, manufacturing company B

4.2.5.1 Key performance indicators

Several of the organisations had KPI's linked to the systematic work environment. Although the most mentioned KPI related to work environment was the lagging KPI for *amount of sick leave days*, some organisations also had proactive, steering KPI's such as *ergonomic introductory trainings/year*. These kind of steering KPI's in ergonomics were considered helpful in guiding the systematic work environment management.

Another lagging KPI described was the T/A, *Tillbud per arbetsskada* (Incidents/Work-related Injuries). This was seen as a usable indicator which was easy to understand and report by employees. In the same organisation, the *Lost Time Injury Frequency Rate (LTIFR)* was also used as a KPI. The LTIFR is a key safety metric used to measure the number of work-related injuries that result in time away from work (Shannon et al., 1996). The interviewees expressed skepticism about using LTIFR as a KPI, since they considered it should be used more for monitoring trends and not for goal setting. The same interviewees also discussed that a problem with LTIFR is that you cannot always be sure if an injury is caused by work.

4.2.6 Ideas for (further) increasing organisational awareness on WEE

A broad variety of ideas and wishes for the future development in the field of WEE were expressed by the interviewees. Greater emphasis on ethical considerations and legal requirements was requested, both at the organisational level and from society more broadly. Interviewees also suggested improvements to WEE tools to better align with their organisational needs.

4.2.6.1 Improvements of WEE tools

The suggested improvements to WEE tools are closely aligned with the previously mentioned motives for their limited use today. On the interviewees wish list are WEE tools that are:

- Evidence based
- Grounded in organisational practices and real-life cases
- Connected to production management
- Easy to understand
- Time-efficient
- Including the whole SWEM process
- Including introductory training in using the tool

4.2.6.2 Organisational level

On the organisational level, some success factors for integrating WEE awareness were identified:

- Leaders' knowledge and prioritization of SWEM
- Support to managers in health care to be able to fulfil their responsibilities
- Company guidelines on ergonomics
- Introductory training in OSH for all employees
- Interprofessional collaboration
- Inhouse OHS who are successful networkers
- Continuous training in OSH and local SWEM responsibilities allocated through the organisation

"I think we really need to reach decision-makers at a higher level and invite them to see our operations firsthand, so they can understand the environment we're talking about. A lot of people up high in the organisation, especially the economists, are so far removed from the reality. They have very little understanding of what the procurement department go through when they choose a slightly cheaper part from a supplier. They see the savings, but they don't realize that the cheaper product might be harder to assemble. So those few pennies or kronor we save might end up being eaten up by other factors."

OSH specialist, manufacturing company A

4.2.6.3 Societal level

On the societal level, the integration of OSH into undergraduate education for engineers and economists was proposed by one interviewee. The idea was that this can support interprofessional collaboration and enhance organisational competence in work environment issues. Interviewees described challenges in helping procurement specialists and designers understand the working conditions of assembly workers and the potential impact of their decisions on those conditions. Integrating OSH into undergraduate education for engineers and economists was seen as potentially contributing to increased trust in the evidence produced within the field of WEE.

"This is an engineering organisation. I really wish engineers got more training in work environment during their engineering education. Same for economists. I think more professional groups would benefit from getting basic knowledge of work environment issues early on. That would have been extremely valuable. Then we wouldn't have such a big need for functions like ours. We need to increase knowledge much earlier, right when people are shaping their careers. Rather than doing it at the workplace. And then we'd get to the point much faster where we start thinking and talking about what improvements need to be made, instead of asking 'Why should I do this?'."

OSH manager, manufacturing company A

5. Discussion

In this chapter, the methods and the results of the literature review and the interview study are discussed and connected to the literature. Future research in the field of WEE is suggested.

5.1 Discussion of the literature review

In this section, the results and methods and limitations of the literature review are discussed. Suggested ideas and improvements for future research are presented.

5.1.1 Discussion of the results of the literature review

The literature review revealed that during recent years, more general business and health economic methods have taken precedence over targeted WEE approaches in the scientific literature on OHS risks and investments. Health economic methods such as CBA, CEA, CUA, FCM and HCM were found multiple times. In contrast to Rose et al. (2013), no methods were identified that integrated both risk assessment and economic evaluation. Only one study (Ramos et al., 2015) combined risk analysis with proposed interventions and a subsequent investment analysis. One of the nine methods identified in the review by Rose et al. (2013), the Balloon Model (Johansson & Johrén, 2017), was also found in the present study.

Several tools described in the earlier review by Rose et al. (2013) proved difficult to locate in current literature. For example, while the ROHSEI tool is reportedly available for purchase (ROHSEI, n.d.), it was not cited in any recent peer-reviewed publications. Tools such as WEST, ProductAbility, SCA and MAWRIC could not be identified through scientific or general online searches. A publication describing “The Potential” (Bergström, 2005) was located, but no recent follow-up studies were found.

One possible explanation for the limited presence and use of these tools may include a lack of dissemination and accessibility, where many are not available for download or practical application. Language barriers and a lack of scientific documentation further limit their adoption. In addition, some methods may be too complex or time-consuming for integration into routine occupational health and safety (OHS) practice, which would further reduce their practical utility.

5.1.2 Discussion of the methods of the literature review

While the current review provides valuable insights, one notable limitation of the study was that it was conducted by a single reviewer. Involving additional independent reviewers would likely have enhanced the methodological rigour by increasing transparency, reproducibility, and credibility of the review process and findings. Furthermore, the inclusion of multiple reviewers could reduce the risk of unconscious bias, which may otherwise affect study selection, especially when judgment is required to assess relevance. It could also increase the likelihood of identifying all eligible studies, particularly in cases where abstracts are ambiguous or where methodological nuances require interpretation. It should also be acknowledged that the selected search strategies (e.g., specific search terms and databases)

may have shaped the findings, insofar as particular economic methods might inadvertently have been excluded because of these choices and therefore not represented in the retrieved literature.

Given the evolving and interdisciplinary nature of WEE, it is also important to consider expanding database selection or including grey literature repositories more systematically. This would help capture emerging tools and methods that may not yet be widely cited in peer-reviewed journals but are nevertheless used in practice.

5.2 Discussion of the interview study

In this section, the results, methods and limitations of the interview study are discussed. Suggested ideas and improvements for future research are presented.

5.2.1 Discussion of the results of the interview study

The interview findings underscore that a one-size-fits-all approach is unsuitable for WEE methods. Interviewees emphasized that for tools to be adopted, they must be both practically usable and trustworthy. These requirements highlight the importance of grounding future tool development in real-world contexts, preferably through participatory and usability-focused design processes. In the interviews, both ethical aspects of WEE as well as alternative approaches to WEE were highlighted.

Differences across sectors were evident. In manufacturing and logistics, production and quality data are readily available and often used. In contrast, sectors like healthcare and education struggle with quantifying work environment impacts due to a lack of standardised data and clear financial metrics. Well-known sector-specific challenges were also confirmed: Musculoskeletal disorders dominate in manual labour sectors, while organisational and psychosocial risks are more prominent in healthcare and white-collar roles, however digitalisation was increasingly creating cognitive demands across all sectors.

Interviewees expressed a dual demand for simplicity and validity in WEE tools. This tension mirrors concerns raised in previous literature, including Rose & Orrenius (2007) and Abrahamsson (2020), particularly regarding the limited availability of reliable, actionable data to support calculations.

Several interviewees stressed that leadership and professional networks are crucial enablers of proactive work environment management. Yet, SWEM is often fragmented (Åkerström et al., 2025). The insufficient managerial support in healthcare that was raised by one interviewee is confirmed by Åkerström et al. (2025) who state that SWEM can itself become a work environment burden for healthcare managers.

A recurring theme in the interviews was the disconnection between SWEM and business management structures. Financial, productivity and work environmental concerns are frequently managed in silos, with strategic and operational efforts running on parallel tracks. This lack of integration undermines organisational learning and long-term planning. Aligning work environment efforts with core business goals was highlighted by all participating organisations as desirable but not always realised. Interestingly, WEE tools were considered most useful in organisations with lower SWEM maturity, where economic arguments are needed to justify interventions. In contrast, in more mature organisations where ergonomics

and SWEM are integrated into daily operations, financial tools played a lesser role. This aligns with the concept of ergonomics infrastructure (Berlin, 2011), which describes the systemic conditions that either enable or hinder the organisational integration of ergonomics into planning, design, and strategic decisions. Two organisations in this study demonstrated strong ergonomics infrastructure, which facilitated proactive work environment efforts and reduced reliance on financial justifications.

Interviewees in organisations with a weaker ergonomics infrastructure highlighted the need to address the knowledge gap in ergonomics within their organisation, for instance by drawing on evidence such as the study by De Neve et al. (2023), which provides compelling support for the link between employee well-being and firm performance. However, Johanson and Aboagye (2020) argue that merely raising awareness of ergonomics is insufficient; meaningful change requires embedding occupational health into business governance systems—a view echoed by the European Agency for Safety and Health at Work (2010).

Although larger organisations are typically more likely to implement SWEM, the five large organisations in this study still reported several barriers. It is therefore reasonable to assume that smaller organisations may face even greater challenges, both in terms of capacity and resource allocation.

A recurring theme in the interviews was the ethical complexity of applying economic methods to human health and safety. While WEE provides valuable tools for illustrating the financial consequences of poor working conditions, there is an inherent risk that workers may be reduced to cost units if such models are applied without ethical safeguards. As highlighted by Johanson & Johrén (2017), this raises important normative questions: Can a monetary value be assigned to a healthy worker? And is it ethically defensible to quantify the costs of work-related injuries and human suffering?

To avoid the instrumentalization of workers, WEE must be grounded in a strong personnel policy that prioritises human dignity and rights (Johanson & Johrén, 2017). As Johanson & Aboagye (2025) state: Calculations are neither neutral nor objective, and they depend on the aim of the individual performing them. Johanson & Aboagye (ibid) present what they call a Pluralistic Economy, where multiple views of economy need to be considered when it comes to issues of sustainability, such as OSH. The aim of calculations should not be to maximize productivity at any cost, but rather to demonstrate that investments in employee well-being are both ethically justified and economically sound (Johanson & Aboagye, 2020). When framed this way, WEE becomes a tool not only for financial evaluation, but also for accountability, reinforcing the societal and organisational responsibility to provide safe and sustainable working conditions. Handled with care, WEE can support a more holistic and human-centred approach to occupational health, which integrates ethical considerations into the language of business and decision-making, rather than setting them in opposition to each other (Johanson & Johrén, 2017). Figure 5 presents a model for future WEE integration into business control. The model draws on the connected themes emerging from the content analysis of the interviews, connected with Johanson & Aboagye's (2020) proposed framework for integrated business control. The model suggests that, for the successful integration of WEE into business control, greater organisational awareness is required. Such awareness calls for both new WEE tools and alternative approaches to account for WEE

effects. The effective development of such WEE tools, in turn, rests on three key considerations to ensure credibility, usability, and local relevance: the main cost factors associated with poor OSH in the organisation, the reasons for the current non-use of WEE tools, and the potential drivers for their future adoption. The integration of business and OSH management was found to be one of the success factors for organisational WEE awareness in the interview study results.

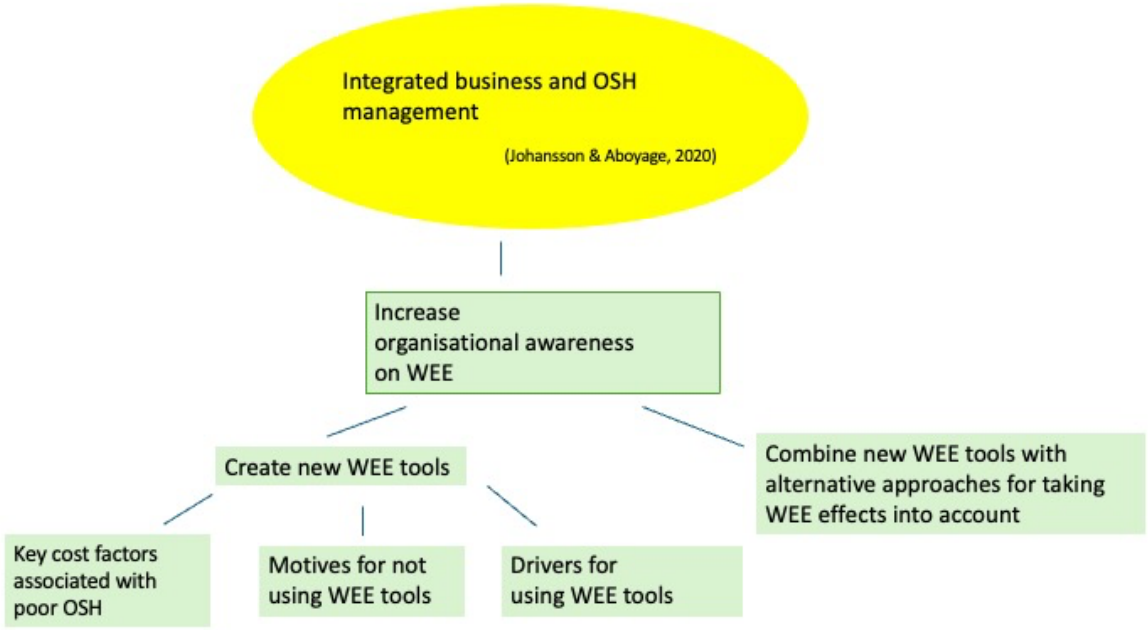


Figure 5: *A model for future WEE integration into business control where the proposed framework for integrated business control by Johanson & Aboagye's (2020) is connected to the themes from the current interview study.*

5.2.2 Discussion of the methods of the interview study

The interview study was based on convenience sampling through the author's and supervisor's networks, which likely resulted in a sample of organisations with relatively high awareness of systematic work environment management (SWEM). Therefore, the findings should not be seen as representative of Swedish organisations at large. A self-selection bias may also be present, as participants were likely those with a strong personal interest in work environment issues, which could influence the nature of their responses.

Interviews were conducted both via Zoom and in person. While the use of Zoom enabled participation from geographically dispersed organisations, it may have limited the ability to interpret non-verbal cues and build rapport, which can affect conversational depth.

Although credibility was strengthened through interviewee validation and peer debriefing, the analysis could have been further validated by involving an additional coder. Lastly, the small and somewhat homogeneous sample limits the transferability of the findings. Future studies should aim for broader sampling across sectors and SWEM maturity levels, e.g. by including also smaller enterprises.

5.3 Synthesising the literature review and the interview study

The use of WEE methods was rare among the participating organisations. None of the tools identified in the 2013 review by Rose et al., nor those highlighted in the current literature review, were referenced by the interviewees. Instead, most participants were familiar with the Mynak tools or internally developed methods, typically sourced from grey literature rather than peer-reviewed research.

Interviewees consistently emphasised the need for simplicity and practical relevance, which aligns with previous findings (Rose & Orrenius, 2007). This suggests that existing tools may be perceived as overly complex or time-consuming, with limited perceived return on time investment for daily operations. However, the balancing act between simple yet accurate tools is a challenge for future developers.

An interesting theme from the interviews was that the identified knowledge gap in WEE concerned primarily the broader organisation, not the OSH professionals themselves. However, the possibility remains that ergonomists and OSH practitioners may also lack formal training in economics or organisational governance. This could also be the reason for the discrepancy between the economic methods described in research and those mentioned by the interviewees. Knowledge gaps in economics and organisational governance could hinder effective cross-disciplinary collaboration. This highlights the need for cross-sector education, where students in engineering and economics learn about SWEM and ergonomics, and future OSH professionals learn about economics. At KTH Royal Institute of Technology, Work environment economics is already a 7.5 ECTS course in the ergonomics curriculum, but further emphasis on economic evaluation in training of future OSH professionals could help bridge the gap between OSH and business decision-making.

In relation to the Sustainable Development Goals (SDGs), the themes of Good Health and Well-being (Goal 3) and Decent Work and Economic Growth (Goal 8) were strongly reflected in the interviews. Discussions often centred around risk reduction (Goal 3d) and the importance of creating sustainable, high-quality jobs (Goal 8.3). While the ethical dimension of WEE was acknowledged by several participants, the gender perspective, also central to the SDGs, was notably absent from the interviews and represents a potential area for future research and practice.

5.4 Practical implications

A key insight from the interview study was the potential for improving OSH by integrating work environment considerations into broader business strategy and governance structures. Several participants noted that when work environment issues are linked to core business goals—such as productivity, quality, or employee retention—they gain visibility and legitimacy at the strategic level. This integration facilitates more systematic and proactive management of OSH.

Traditionally, OSH efforts have been reactive, often focused on complying with regulations or addressing incidents after they occur. However, aligning OSH with strategic planning, performance measurement (e.g., through leading indicators), and investment analysis can shift the focus toward prevention and long-term improvement. This observation echoes earlier findings by Berlin & Adams (2017) and Pawlowska (2015), who noted that high-performing organisations are more likely to use proactive and forward-looking metrics in their OSH work.

When WEE is integrated into governance, it enables OSH professionals to engage more effectively in business-critical conversations. Rather than advocating for safety solely on moral or legal grounds, they can also present the economic case for healthy workplaces, connecting employee well-being to performance, innovation, and sustainable growth.

This study included only five, but nevertheless large organisations. The findings contribute with valuable implications for future developers of WEE methods as well as an opportunity for reflection for the particpating organisations. The current overview of the field of WEE can possibly benefit future students in the field of OHS, engineering and economics.

6. Conclusions

The aim of this thesis was to contribute to the understanding of WEE by presenting a review of recent research in the field and describing how a selection of Swedish organisations in the manufacturing, logistics, healthcare and higher education sectors take into account and address the financial effects of their work environment.

The literature review, covering research published between 2012 and 2024, revealed that more general business and health economic methods have taken precedence over targeted WEE approaches in the scientific literature on OHS risks and investments during recent years.

The interview study revealed that while awareness of WEE tools was high, actual usage was rare. Two out of five participating organisations reported that the use of economic methods had facilitated their work environment management. Participants expressed a clear need for simpler yet credible tools that are tailored to organisational realities and easily integrated into local business contexts. Alternative approaches to WEE included proactivity and integration of SWEM into business operations.

Motives for taking WEE effects into account included OSH professionals raising awareness among management about the benefits of SWEM interventions and occupational health interventions. The findings suggest that WEE calculations may be especially useful in organisations with a lower maturity in SWEM, where financial arguments are more critical for justifying interventions. In contrast, organisations with a strong integration of SWEM into core operations rely less on formal economic arguments, as the value of a healthy work environment is already embedded in their culture and organisational goals.

Looking ahead, based on the results of the interview study, the development of WEE tools should prioritise participatory design and practical usability to ensure relevance and effectiveness in real-world settings. Such an approach can support both strategic decision-making and ethical, sustainable investments in employee health and well-being.

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Appendix A: Included studies in the literature review, categories 1-3.

***Table A1**. Included original articles for category 1: Identification of WE risks and calculation of their economic impact.*

Author	Country	Method(s) used for calculation of WE risk/cost factor	Type of risk or cost investigated	Study design	No of subjects	Main findings
Cho et al. (2023)	USA	A nurse staffing model that incorporates patient length-of-stay, nurse turnover, and costs related to patient-to-nurse ratios.	Patient length-of stay, Nurse turnover	Presentation of a model	3 hospitals	Lower patient-to-nurse ratios can potentially provide hospitals with financial benefits in addition to improving the quality of care. Higher policy patient-to-nurse ratio upper limits may not be as harmful in smaller hospitals, but lower policy patient-to-nurse ratios may be necessary for larger hospitals. The ratio should be tbe hospital dependent.
Lui et al. (2022)	Hong Kong	The Human Capital Method (HCM).	Presenteeism Productivity loss costs (presenteeism + absenteeism)	Survey	4500 nurses in 3 hospitals	Presenteeism prevalence is high amongst acute-care hospital nurses and workload increase during influenza flu surge significantly heightened nurse sickness presenteeism. Annual presenteeism productivity loss costs in this study of USD 24,096 were one of the highest reported worldwide.

Author	Country	Method(s) used for calculation of WE risk/cost factor	Type of risk or cost investigated	Study design	No of subjects	Main findings
Nagata et al. (2022)	Japan	The human capital method (HCM)	Costs of medical and dental treatment Costs of presenteeism	Cross-sectional study	3910 workers in a pharmaceutical company	Costs associated with presenteeism and medical treatment among high-stress workers were higher than the costs among non-high-stress workers, whereas the costs associated with dental treatment were not. It is recommended that presenteeism measurement items be added to the stress check survey, and that the methods used in the study is used to calculate the associated losses although with careful interpretation.
Popescu-Stelea et al. (2019)	Romania	Systematic Accident Cost Analysis Method (SACA)	Costs related to an acciddent	Review of models for workplace accidents and illnesses cost analysis tools	-N/A	A simple approach to cost classification. It requires strict definitions of the cost categories used to ensure validity. Only used in Danish companies.

Author	Country	Method(s) used for calculation of WE risk/cost factor	Type of risk or cost investigated	Study design	No of subjects	Main findings
Russo et al. (2021)	Italy	A cost estimation model for work-related stress, based on the Human Capital Approach (HCA). The population attributable fraction (PAF) epidemiological measure was used to calculate the impact of exposure to work-related stress based on data collected through validated national tools and salary cost data.	Loss of productivity due to days of absence attributable to work-related stress	Two case studies	Case study 1: Public hospital workers (N=1014) Case study 2: Public administration department (N=534)	The developed model was implemented and evaluated in the two organisations. In the first case (healthcare), it was found that absence related to work-related stress cost the organisation €445 000. In the second case (administration), the cost was €360 000.

Author	Country	Method(s) used for calculation of WE risk/cost factor	Type of risk or cost investigated	Study design	No of subjects	Main findings
Strömberg et al. (2017)	Sweden	Wage multipliers that can be used by managers to estimate the costs of productivity loss for employers in economic evaluations.	Costs for lost productivity due to absenteeism, acute presenteeism, chronic presenteeism, and problems related to the work environment.	Cross-sectional survey	758 managers from different sectors of the labor market.	Job characteristics were linked to productivity loss because of health-related and work environment– related problems. The impact of impaired performance on productivity differed among occupations. The mean wage multiplier was 1.97 for absenteeism, 1.70 for acute presenteeism, 1.54 for chronic presenteeism, and 1.72 for problems related to the work environment. This implies that the costs of health-related and work environment–related problems to organisations can exceed the worker's wage.
Vänni et al. (2017)	Finland	A model for calculating the total loss in days due to presenteeism at the company.	Absenteeism Presenteeism	Cross-sectional. .	A company within food industry: Four factories and one administrative unit. N=2700.	The share of presenteeism costs was constant at about 1% of annual turnover and about 3.7% of personnel costs. The lowest annual presenteeism cost per employee was EUR 986 and the highest was EUR 1302. The lowest number of presenteeism days per employee in a year was 8.7 days and the highest number was 10.4 days. Estimated losses to a company due to sickness absences and presenteeism ranged from EUR 4.6 million to EUR 5.6 million annually.

Table A 2. *Included original articles for category 2: Development and evaluation of suggestions for intervening measures in combination with investment analysis.*

Author	**Country**	**Method(s) used for development of suggestion for Intervening Measure and Investment Analysis**	**Type of intervening Measures**	**Study design**	**No of subjects**	**Main findnings**
Ramos et al. (2015)	Portugal	Cost-benefit analysis (financial) Cost-benefit analysis (economic)	A risk analysis of the work accidents in a hospital and investment analysis using Benefit/Cost ratio (B/C) in financial terms (from the organisation's perspective) and in economic terms (including the benefits for the worker and for the Society).	Case study	Hospital with 2500 employees	The financial ratio was only advantageous in some cases. When the societal perspective was considered, the B/C ratio increased significantly.

Table A 3. *Included original articles for category 3: Investment analysis in connection with work environment (WE) investments.*

Author	Country	Method(s) used to calculate financial impact of WE investments	Type of investment	Study design	No of subjects	Main findings
Dubois et al. (2020)	Canada	Cost-benefit Analysis	Multidimensional intervention that contributes to developing resilience of interdisciplinary cancer care teams and thereby reduce mental health problems.	Study protocol for a planned longitudinal quasi-experiment al design	In total 700 workers, from four institutions.	N/A
Severin et al. (2022)	Sweden	Cost-benefit analysis	Several different organisational interventions for reducing absenteeism	Cohort study	4802 in total, divided in two different intervention groups	This program was economically beneficial in terms of reducing the productivity loss, but not of reducing direct sick pay costs connected to short-term sickness absence.
van Dongen et al. (2013)	The Netherlands	Cost-effectiveness Analysis from societal perspective. Return on Investment Analysis from company perspective.	A Worksite Vitality Intervention among Older Hospital Workers including Yoga and aerobic exercising, coaching, and fruit.	RCT	730 hospital workers	No significant differences in costs and effects were observed between intervention vs control group. Per euro invested, €2.21 was lost. The intervention was neither cost-effective nor cost saving.

Michaud et al. (2022)	United States	CEA ROI (benefit-cost ratio) Sensitivity analysis	Workplace intervention designed to accompany sit-stand work stations to reduce sedentary time and increase Light-intensity Physical Activity.	Retrospective analysis of a cluster randomized trial	630 participants in 24 worksites	The cost was $1 for each additional minute reduction of workplace sitting time. There was a modest and non-significant change of loss of work productivity improvement which was associated with a $0.34 return for every $1 invested.
(Goggins) n.d.	USA	Washington State Ergonomics Cost Benefit Calculator (WSECBC)	Ergonomics programs and individual health and safety solutions.	The tool is based on a review by Oxenburgh, 1991.	250 case studies	N/A
Noben et al. (2015)	The Netherlands	Cost-benefit Analysis	A preventive screening intervention for burnout, anxiety and depressive disorders. Screen-positive nurses received personalized feedback and referral to the occupational physician.	RCT	413 nurses in a university hospital.	The return-on-investment was 5-11 euros for every euro invested. Within half a year, the cost of offering the preventive intervention was more than recouped.

Braun et al. (2014)	UK	Cost-benefit analysis	Program for work environment changes combined with lifestyle changes.	Retrospective	232 workplaces	Participation was associated with a mean reduction in sickness absence of 0.26–1.6 days per employee per year. The estimated cost for the programme was £3 per sickness-absence day saved.
Gao et al. (2018)	Australia	Within trial Incremental cost- efficacy Analysis according to a pre-specified protocol. Long- term CEA	Intervention to reduce sitting time among desk workers.	RCT	231 participants from 14 workplaces within the same organisation	The intervention was cost-effective over the lifetime of the cohort when scaled up to the national workforce.
Jensen et al. (2022)	Sweden	Cost Effectiveness Analysis	Productivity Measurement and Enhancement System (ProMes) intervention to reduce employee work-related stress and enhance work performance.	Prospective cohort study	Employees at a healthcare unit. N=67	ProMes was not cost–effective in terms of work-related stress levels, but the intervention seemed to have increased efficiency in some objective work performance measures, albeit at a relatively high extra cost.

Appendix B: Information to participants in the interview study

Thesis on work environment economics

You are invited to participate in an interview concerning occupational health economics. The study is part of a master's thesis in the Master's Programme in Technology, Work and Health at the Royal Institute of Technology (KTH). The content of the interviews may also be used in future research projects on occupational health economics within the Division of Ergonomics at KTH.

Purpose
The aim of the study is to provide a situational overview of how a selection of Swedish organisations from various sectors take into account the economic effects of the work environment in their organisations..

Method
Individual interviews will be conducted with approximately ten employees from various organisations. Organisations other than NN will also be included in the study.

Why have you been invited?
We would like to include you in the study because you have knowledge of systematic occupational health and safety management and/or economic aspects within your organisation. You will also have the opportunity to suggest one or more colleagues whom you believe would be relevant to include in the study. Your participation is completely voluntary, and you may withdraw at any time without providing any reason. You may also choose to exclude parts of your interview responses from the study if you wish.

What does participation involve?
You will participate in a one-on-one interview conducted by the study coordinator. The interview will be held in a conversational format and, with your consent, will be recorded. Your participation will in no way affect your current work situation. The interview will last approximately 60 minutes and will take place either at your workplace or via Zoom.

Handling of data and personal information
All collected data from the interviews will be protected according to the Personal Data Act. Before data analysis begins, any possibility of linking your name to the interview transcripts will be removed. Only the researchers involved at KTH will have access to your responses. It will not be possible to trace any individual's answers in the compiled results. All collected data will be securely stored. Your employer will not be able to access any individual's responses. As a participant, you have the right to request information from the project leaders about how your results have been registered. You also have the right to request corrections to any inaccurate information.

KTH is responsible for your personal data. According to the EU General Data Protection Regulation (GDPR), you have the right to access your personal data processed in the study free of charge and to request corrections if needed. You may also request that your data be deleted or that the processing of your personal data be restricted. If you wish to access this information, please contact the principal researcher Linda Rose, phone: +46 (0)8-790 94 96. The Data Protection Officer can be reached at dataskyddsombud@kth.se or by phone: +46 (0)8-790 87 52. If you are dissatisfied with the handling of your personal data, you have the right to lodge a complaint with the Swedish Authority for Privacy Protection (Integritetsskyddsmyndigheten), which is the supervisory authority (formerly the Swedish Data Protection Authority, Datainspektionen, as of January 1, 2021).

Responsible for the study:

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Please feel free to reach out if you have any questions!

Appendix C: Informed consent

Consent to participate

I hereby consent to participate in the research project and that the interview in which I take part may be used in the research study.

I am aware that:

- Participation in the study is voluntary
- I may withdraw from the study at any time without giving any reason
- The interview will be recorded and later transcribed
- Quotes from what I say during the interview may be used in the final report, but in such a way that I remain anonymous
- The content of the interview will be used in a thesis on how organisations consider the economic effects of the work environment, and may also be used in future research projects on occupational health economics at the Division of Ergonomics, KTH

.....
Place and date

.....
Participant's signature

.....
Printed name

Appendix D: Interview guide

Briefing

- I'll start by introducing myself.
- I'll briefly explain the project and the purpose of this interview.
- I will be recording the interview and transcribing it afterward.
- The interview will last no more than one hour.
- We can take a break if you need one—just let me know, or I'll check in with you halfway through.
- Do you have any questions before we begin?

1. To start, I'm interested in hearing about your systematic work environment management.

A) Would you like to describe the tools and methods you use in your systematic work environment management?

B) What systems do you work with?

C) What issues would you say are prioritized in your work environment efforts?

D) What different perspectives do you consider when prioritizing work environment issues?

Let's focus a bit more on your role.

E) What does your role in the work environment efforts look like?

F) If you identify an area where investment or prioritization is needed, how do you justify it? (What arguments do you use to gain support?)

G) What other roles, besides yours, are involved in the work environment management?

H) How is the work environment discussed within your organisation? What arguments do others use to motivate investments in the work environment?

I) How do you choose to work with managers in your work environment efforts?

J) Do you address work environment issues in specific forums?

K) Do work environment issues come up in other management systems besides the one for systematic work environment management? In what ways?

L) Would you like to share a bit about how your working methods have developed over time? (What has driven those changes?)

2. Now we'll move on and talk more about the financial effects of the work environment.

A) First, I'm interested in hearing about your experiences.
Have you had any experience using a specific method to evaluate the financial effects of the work environment (on the company's finances)?

B) If yes, what value do you see in doing that?

C) Is any such financial evaluation method currently used here at [COMPANY NAME]?
(For evaluating how the work environment affects the company's finances?)

If the answer to 2C is no:

E1) Why do you think no such method is being used?

E2) What conditions within your organisation do you think would be required to start using such methods?

E3) What requirements would the method/tool itself need to meet in order for you to consider using it within your organisation?

E4) What factors do you think contribute the most to the costs of poor working conditions (if any) within your organisation?

→ Then continue to section 3)

If the answer to 2C is yes:

F1) In what ways?

F2) What prompted you to use this/these method(s)?

F3) What challenges are associated with using this method/these methods?

F4) What aspects make the method(s) easy to use?

F5) In what ways has the use of this method/these methods influenced your work environment efforts?

F6) What conditions do you see as important for applying financial evaluation methods for work environment in your organisation?

F7) What types of cost categories or factors do you primarily consider?

F8) What types of costs have you found to be the highest in your analyses?

F9) When you calculate the costs of an investment to improve the work environment, do you also consider the alternative costs of doing nothing—that is, continuing with "business as usual"?

3) Do you have any other ways of accounting for the financial effects of the work environment that we haven't talked about yet? (For example, KPI's)

4) If you could wish freely – what would you wish for in the field of work environment economics to support your organisation's efforts?

5) Could financial models for evaluating work environment issues be better adapted to your company's needs? If so, how?

Closing questions:

6) That was my last question. Is there anything else you would like to add before we finish?

7) Do you have any questions for me?

8) I'll now summarize a few themes I've picked up during the interview.
Do you think this summary is accurate?

(Turn off recording)

Debriefing

- Thank you very much for participating!
- How did that feel?
- Is there anything you'd like to add?
- Do you have any questions?
- Do you have any suggestions for colleagues within your organisation that I could interview about this topic?
- I'll now conduct more interviews from a total of five different organisations across various industries, transcribe the interviews, and then analyse them using qualitative content analysis. The results will be presented in my thesis in early June.
- Thanks again!

Appendix E: Themes from the qualitative content analysis of the interviews.

